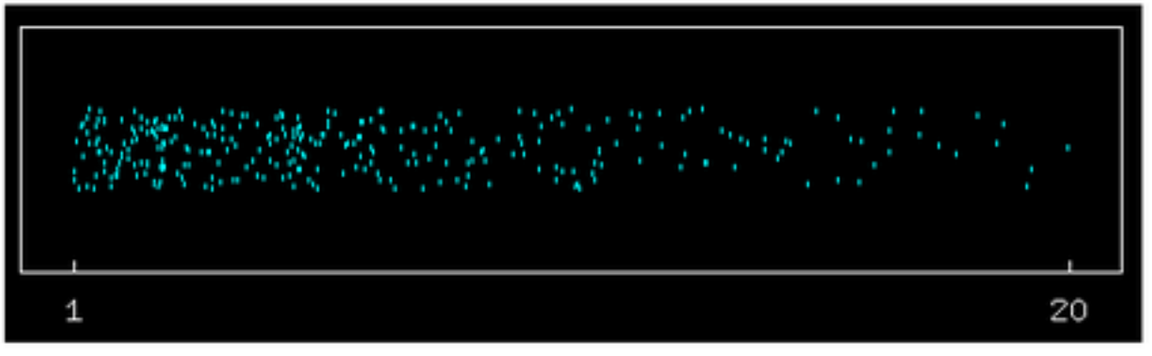
Lecture 10

INFERENCE OF PARAMETERS AND MODELS

Inferring parameters in science experiments

Please think about this question (Exercise 3.3, page 47)

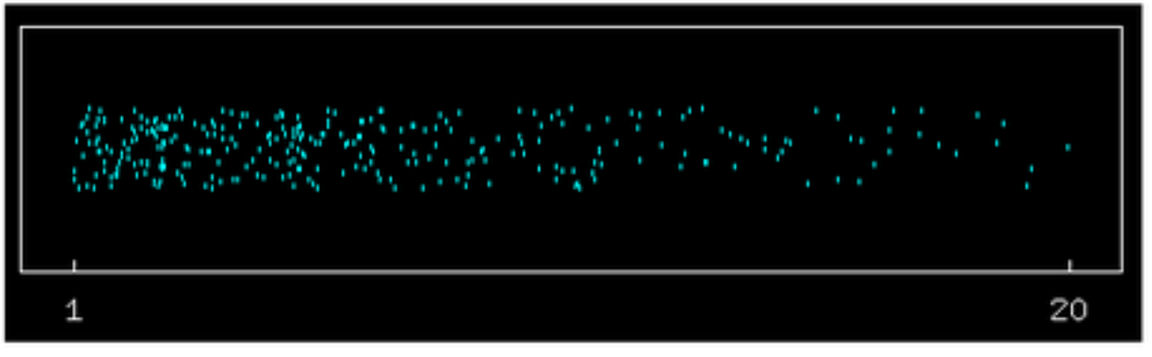


A source emits particles; they decay at locations exponentially distributed with lengthscale λ. Decays can be observed only if the location falls in a window $(x_{\min}, x_{\max}) = (a, b) = (1, 20)$.

N decays are observed at locations $\{x_1, \ldots, x_N\}$. What is λ?

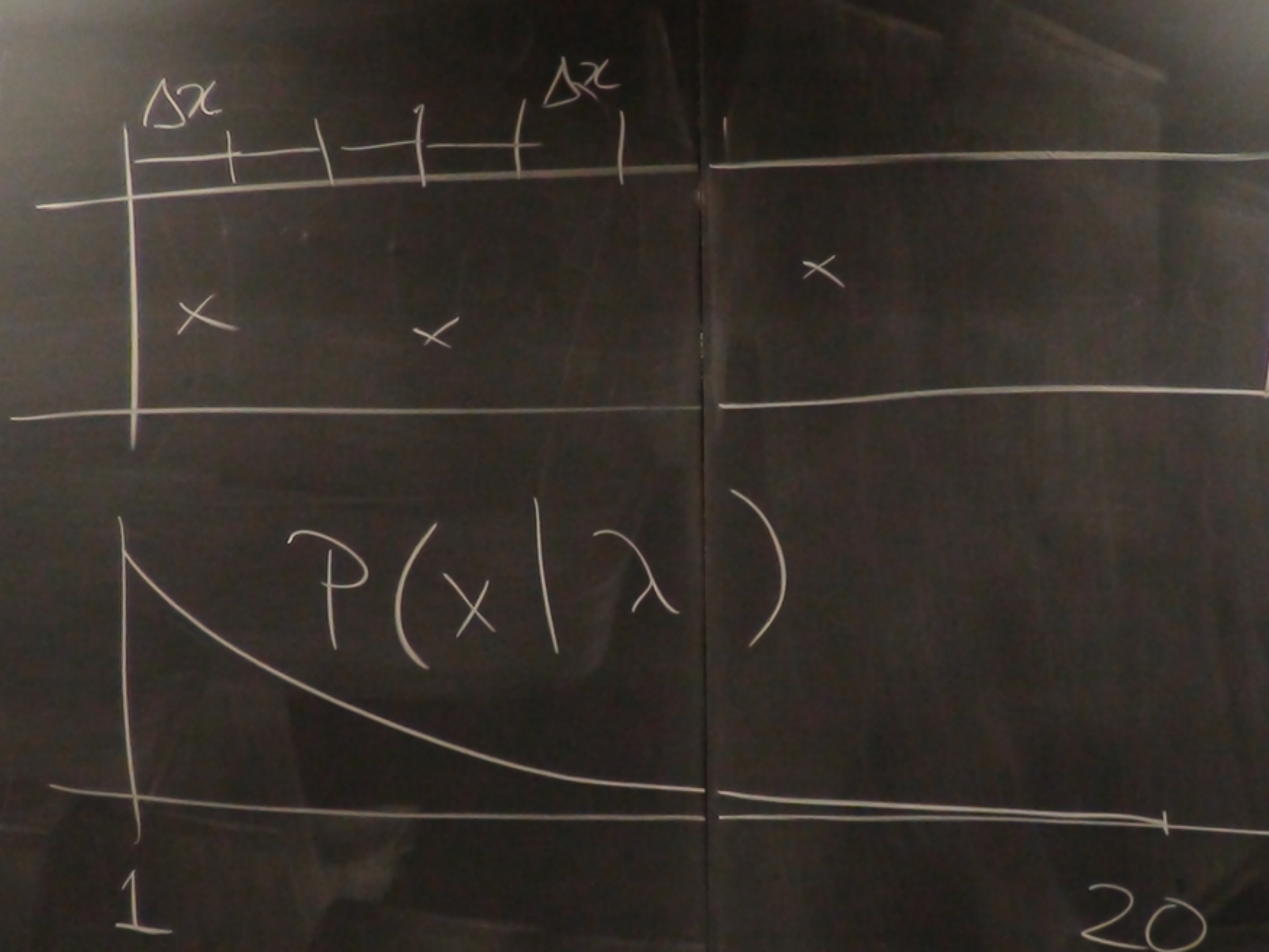
Inferring parameters in science experiments

Q: How would you have solved this problem before you heard about likelihood functions and Bayes' theorem?

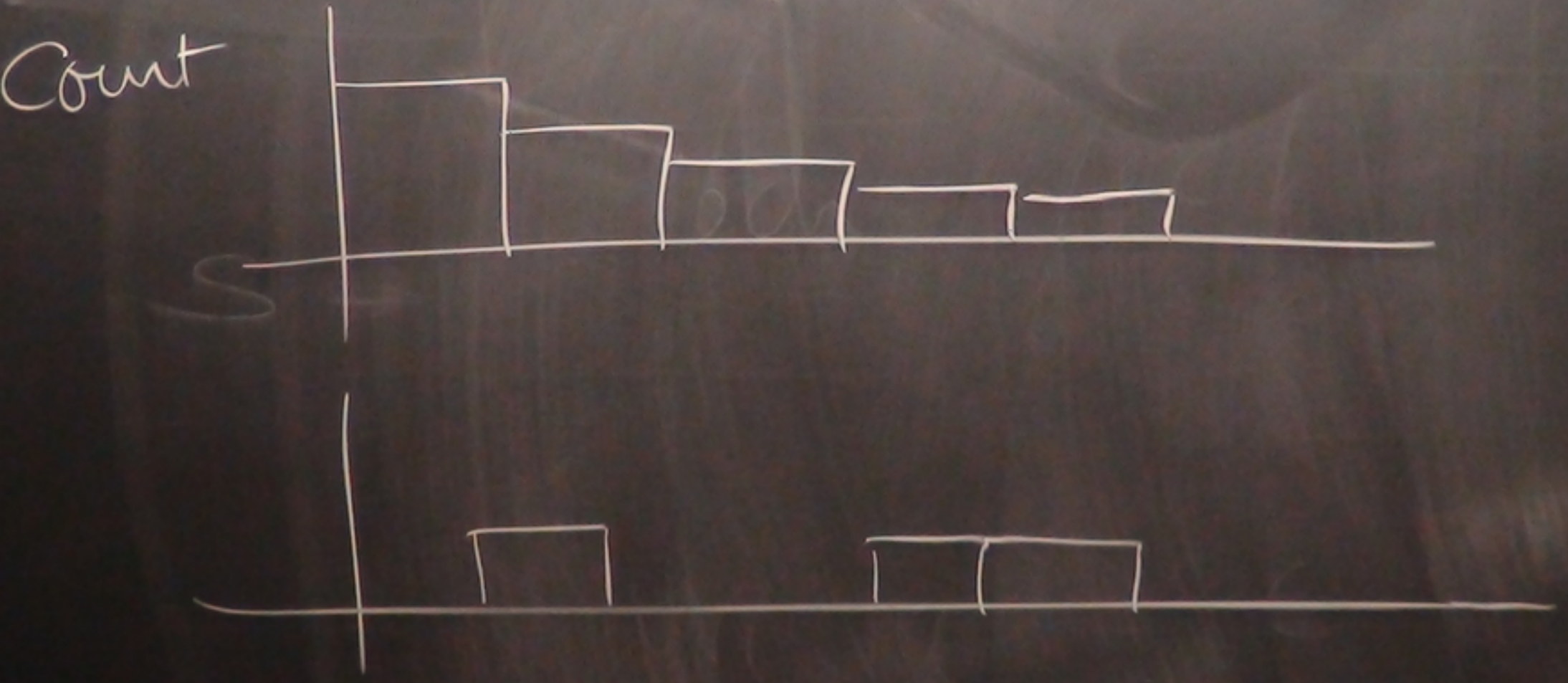


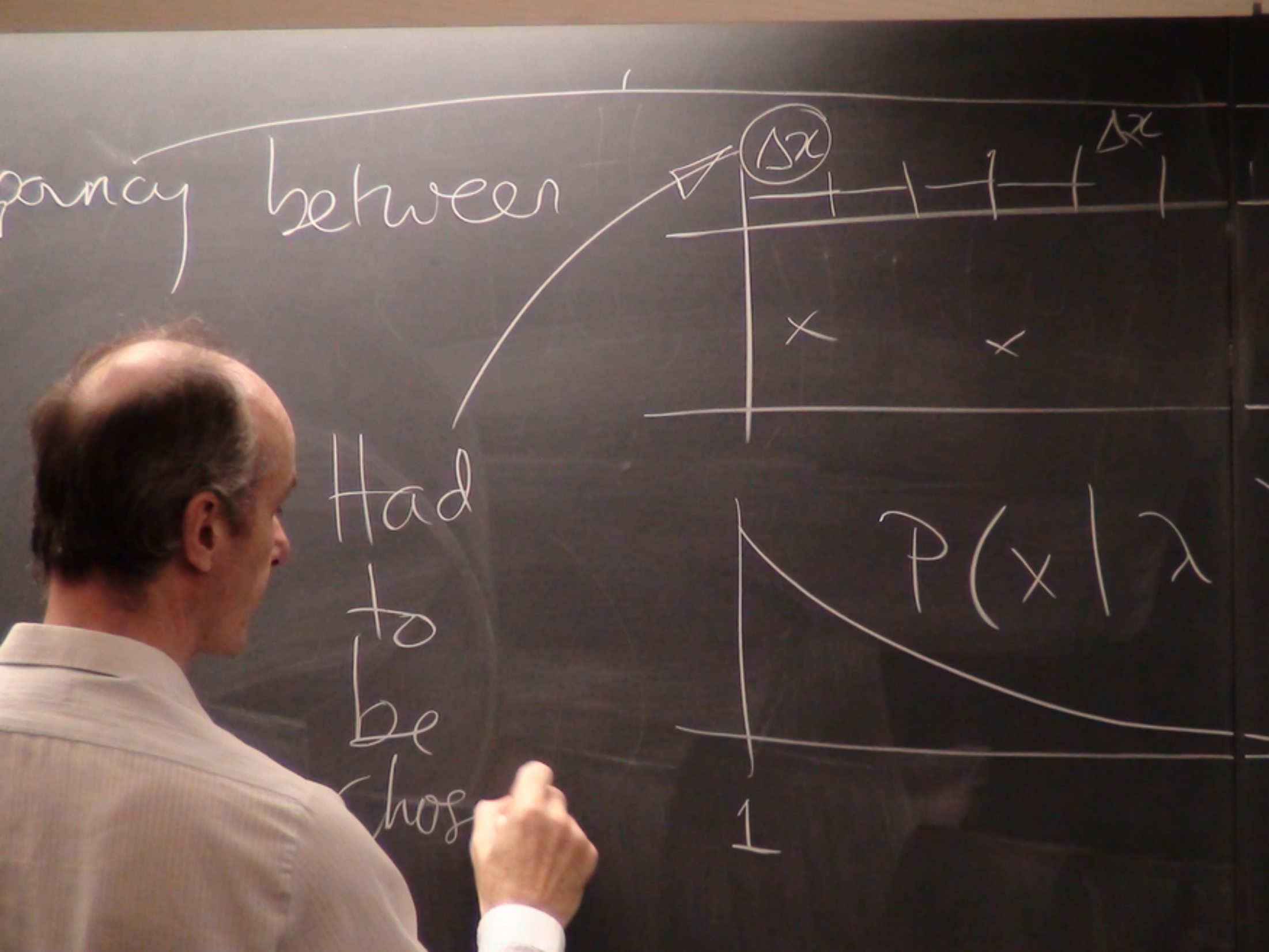
A source emits particles; they decay at locations exponentially distributed with lengthscale λ. Decays can be observed only if the location falls in a window $(x_{\min}, x_{\max}) = (a, b) = (1, 20)$.

N decays are observed at locations $\{x_1, \ldots, x_N\}$. What is λ?


Δx
Δx
x
x
x
P(x|λ)
1
20

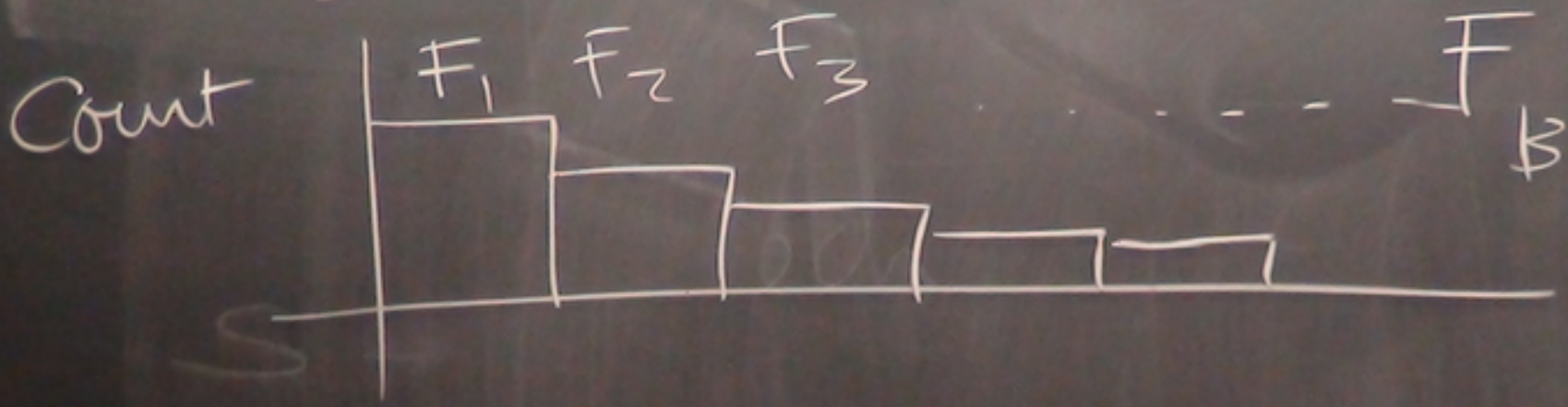
PUT DATA IN BINS

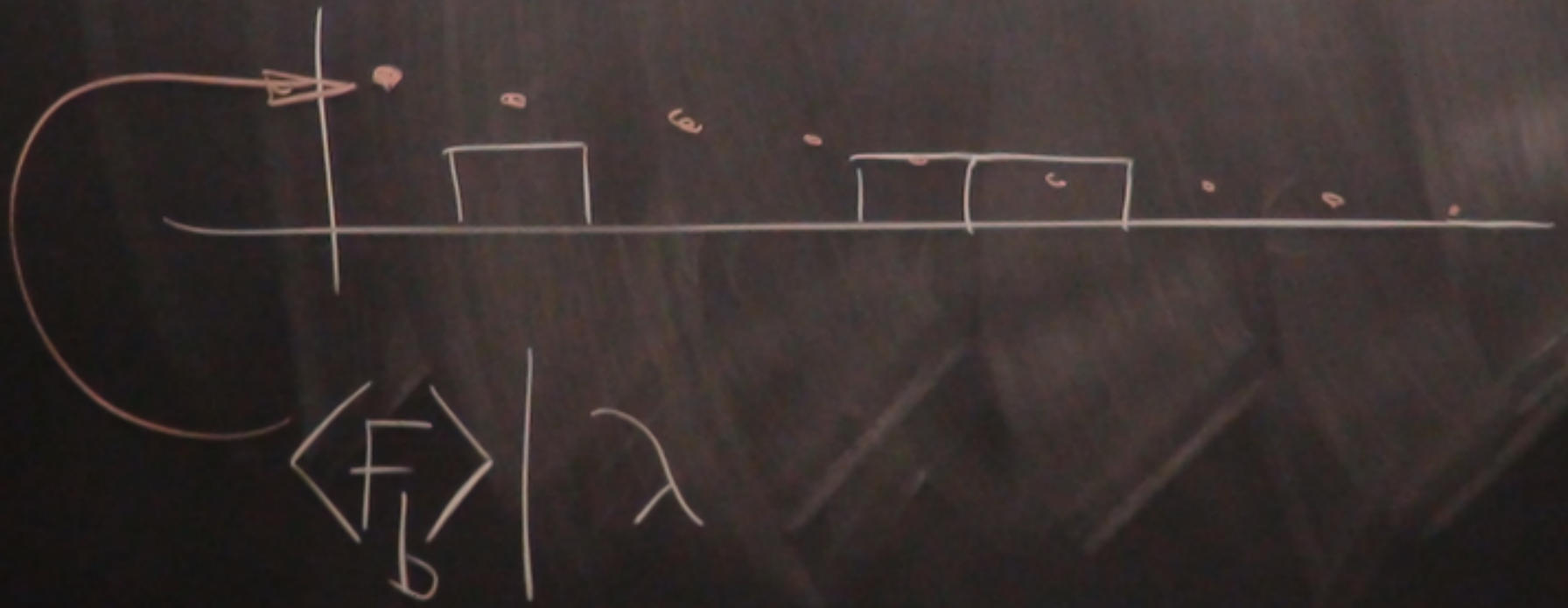



pancy between
Δx
Δx
Had
to
be
chos
P(x|λ
1

PUT DATA IN

BINS





$\langle f_b | \lambda$

$$\text{Goodness of fit}_1 = \sum_b \left(F_b - \langle F_b \rangle \right)^2$$

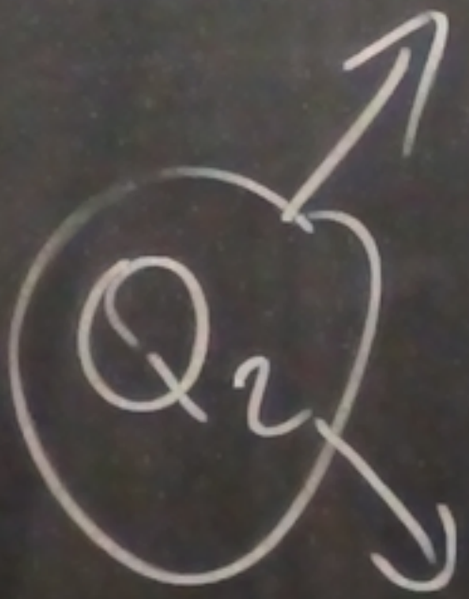
$$\text{Goodness of fit}_2 = \sum_b F_b \cdot \langle F_b \rangle$$

$$\sum_b F_b^2 - 2\sum_b F_b \langle F_b \rangle + \sum_b \langle F_b \rangle^2$$

Constant

hmm?

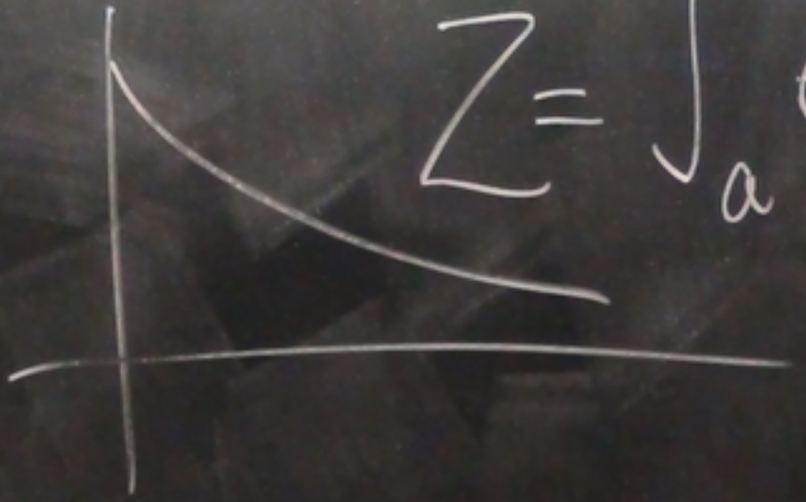
Goodness of fit $_1$ = $\sum_b$



Goodness of fit $_2$ =

$$P(x_n \mid \lambda, a, b) = \begin{cases} 0 & \text{otherwise} \\ \frac{e^{-\frac{x}{\lambda}}}{Z} & x \in (a, b) \end{cases}$$

$$Z = \int_a^b e^{-x/\lambda}$$

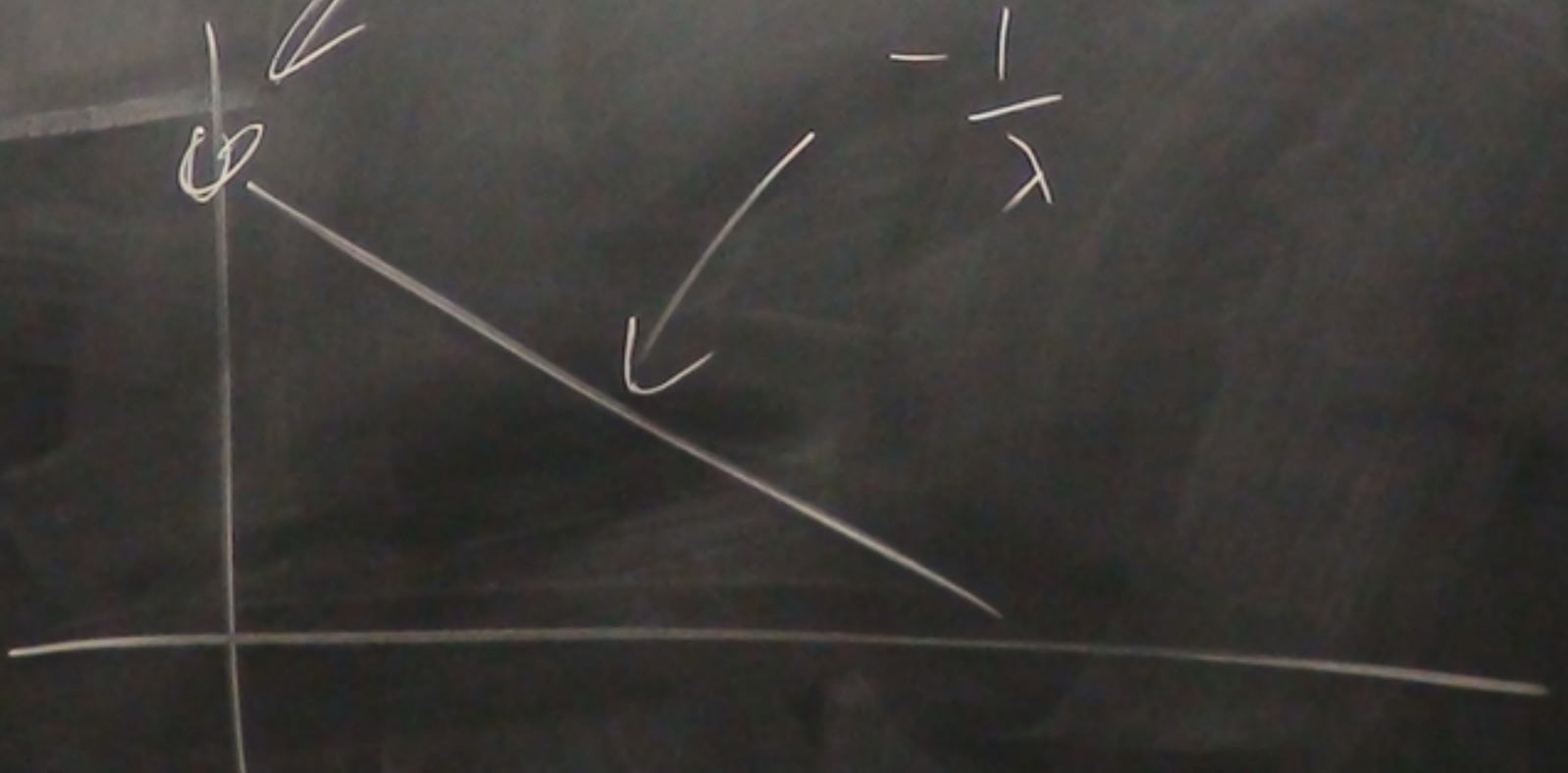


$$P(x|\lambda,a,b) = \frac{e^{-x/\lambda}}{Z(\lambda,a,b)}$$

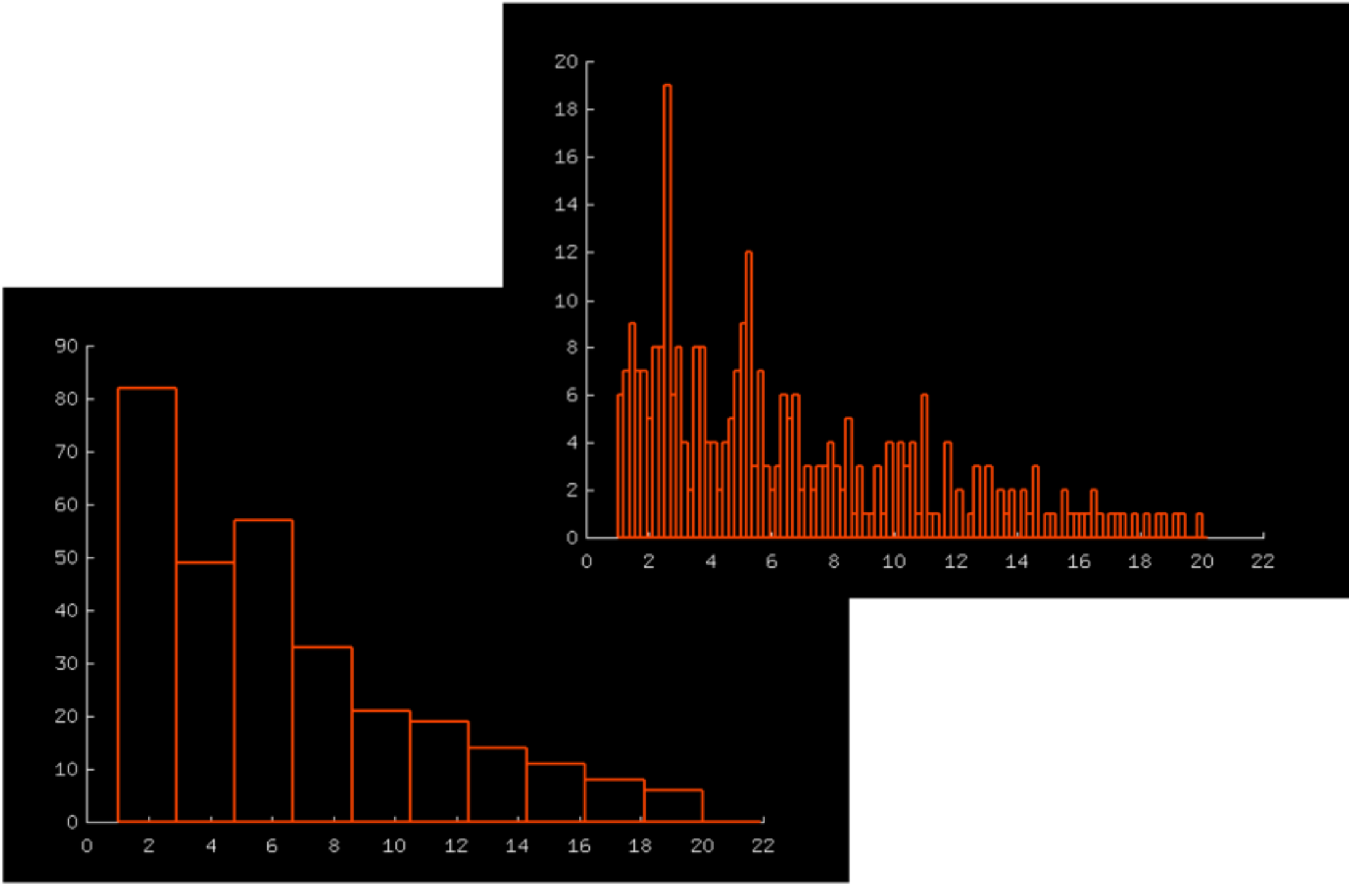
$$\underbrace{NP(x|\lambda,a,b)\Delta x}_{\text{expected number in bin}} = \frac{N}{Z} e^{-x/\lambda}$$

count in bin at $x, \Delta x$

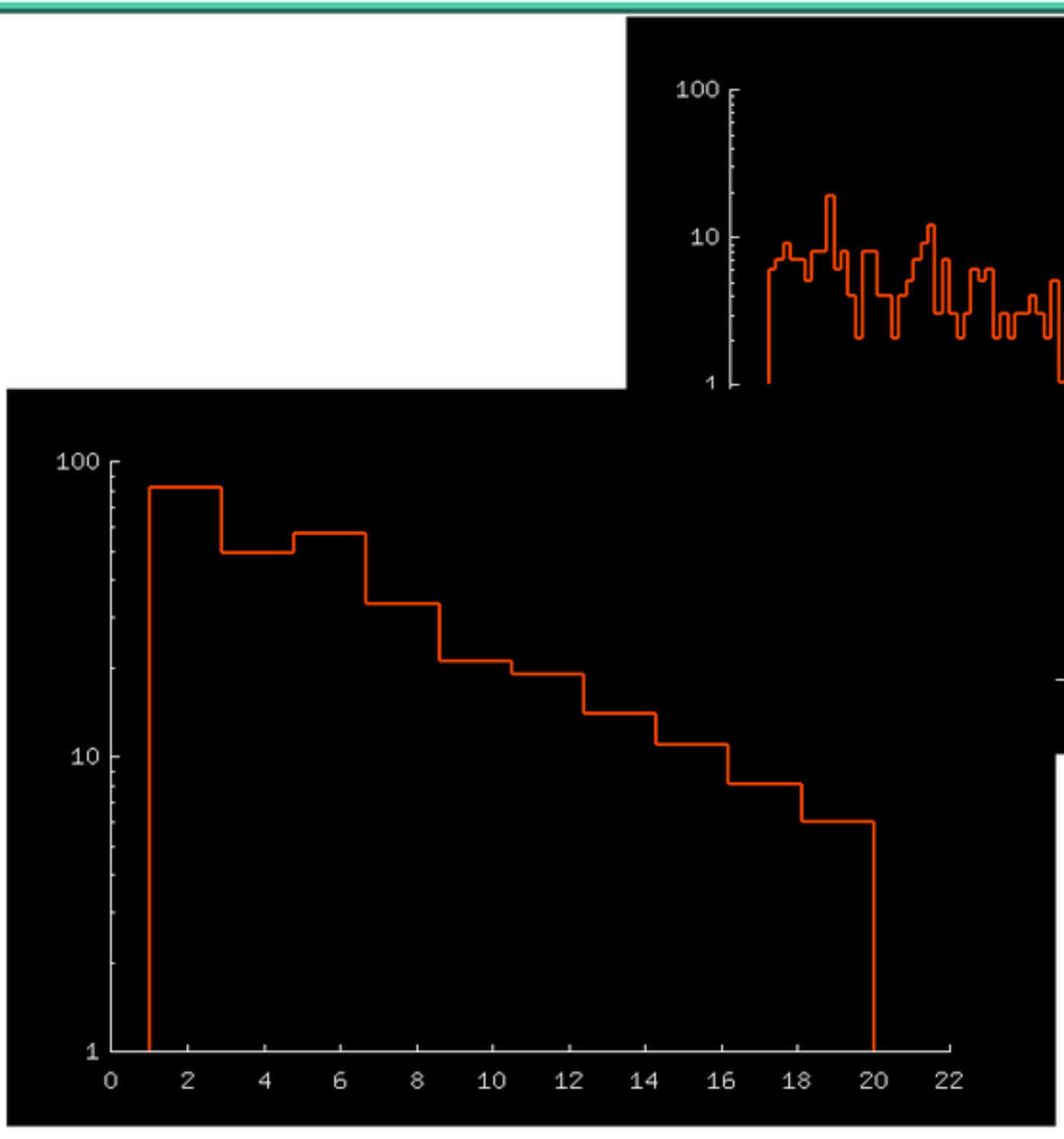
$$\log \langle F \rangle = \log \frac{N}{Z} - \frac{x}{\lambda}$$

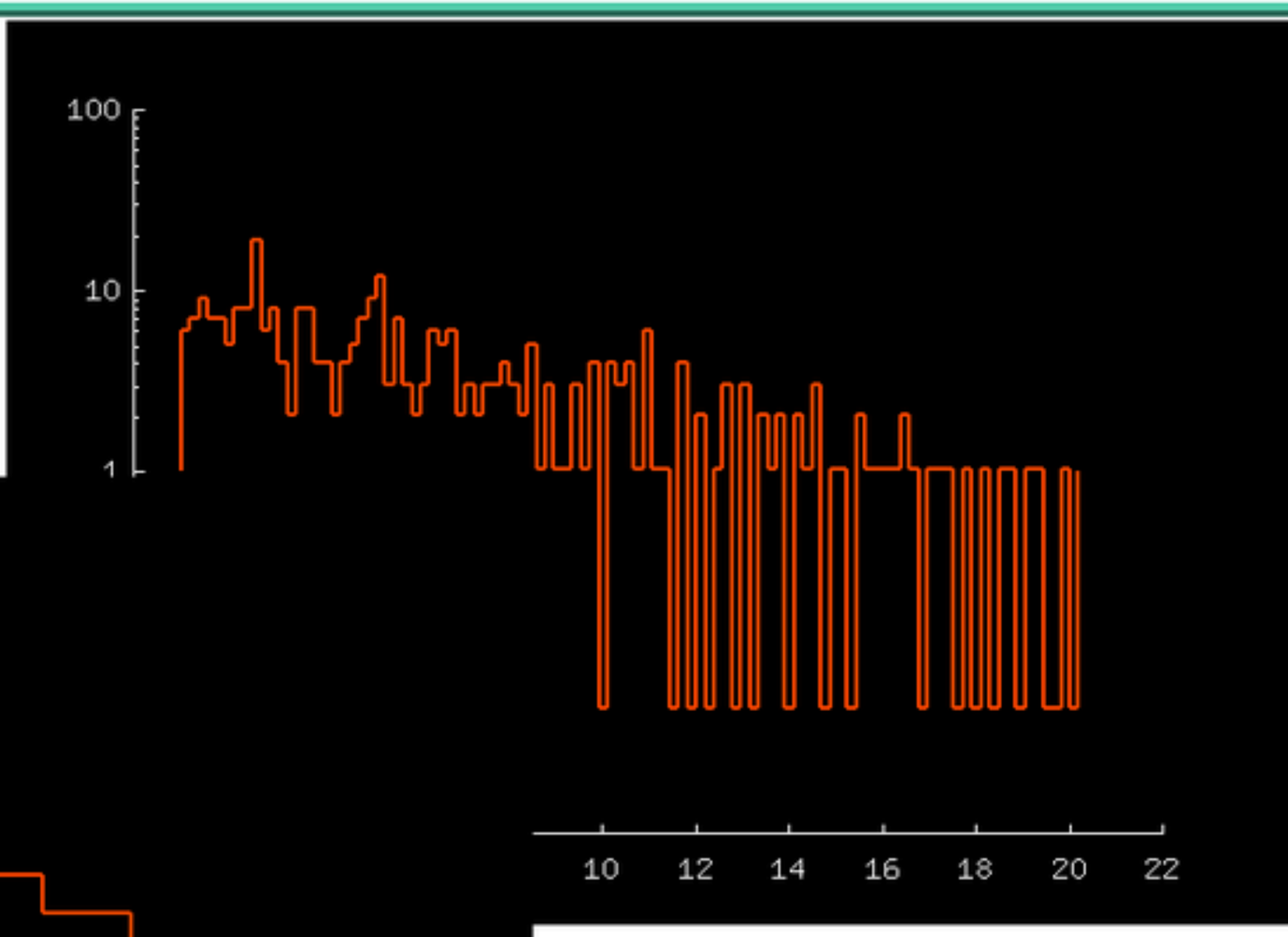


Put data in bins, make histogram

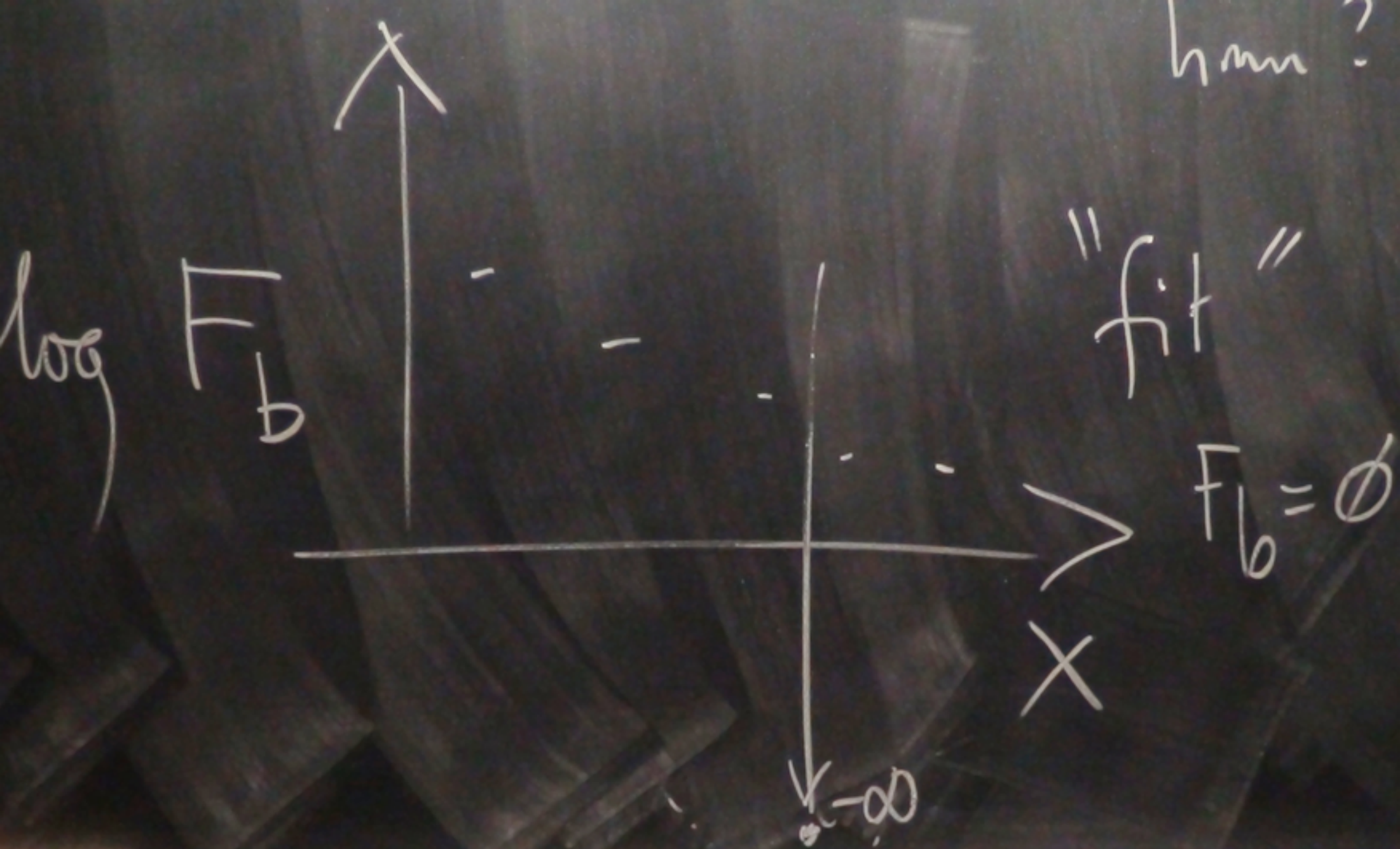


Put data in bins, make histogram, and take log





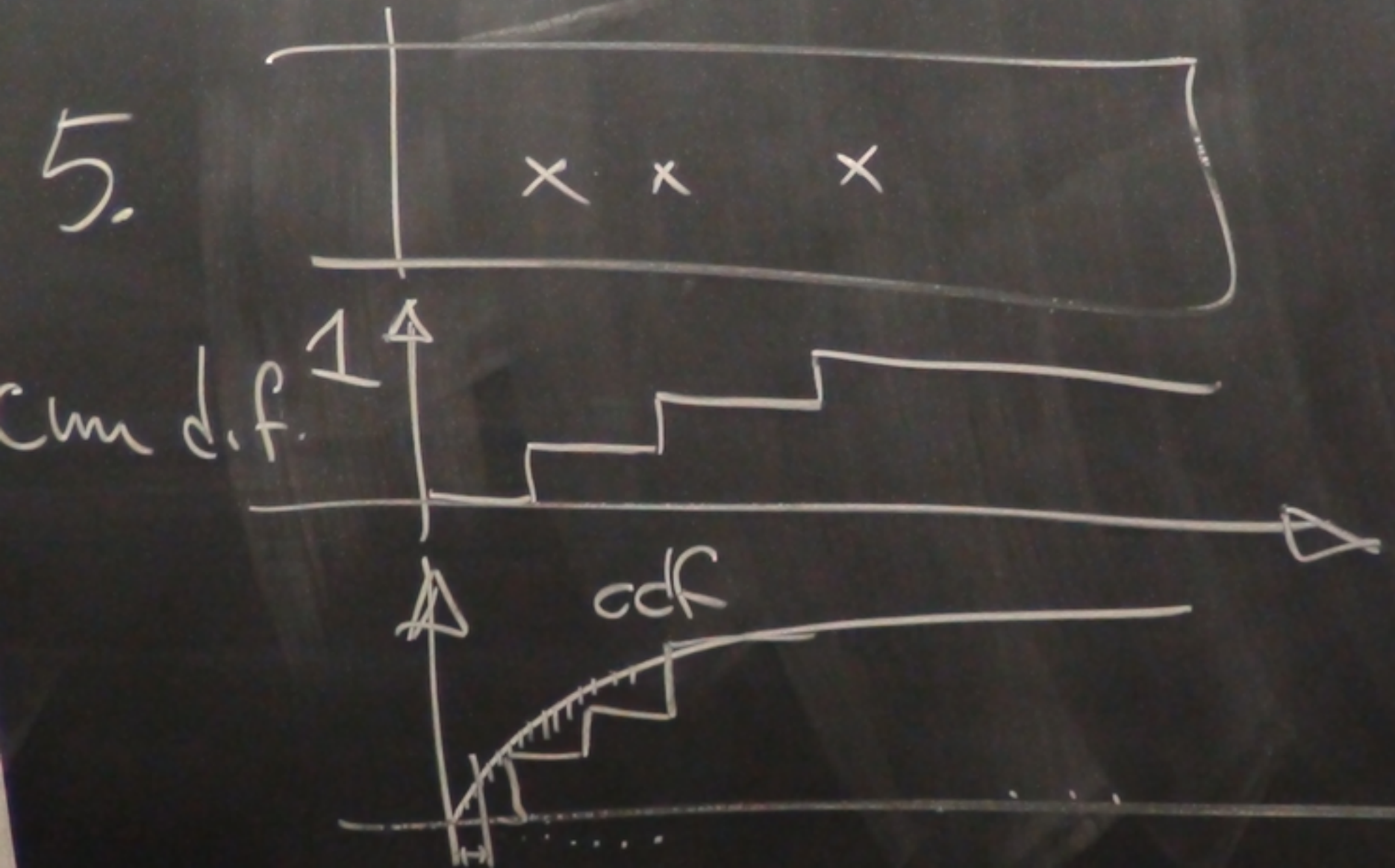
then 'fit a straight line'
(how do we handle
zero counts?)


hmm ?
log F_b
"fit"
$F_b = \emptyset$
X
$-\infty$

pdf

by Gradien

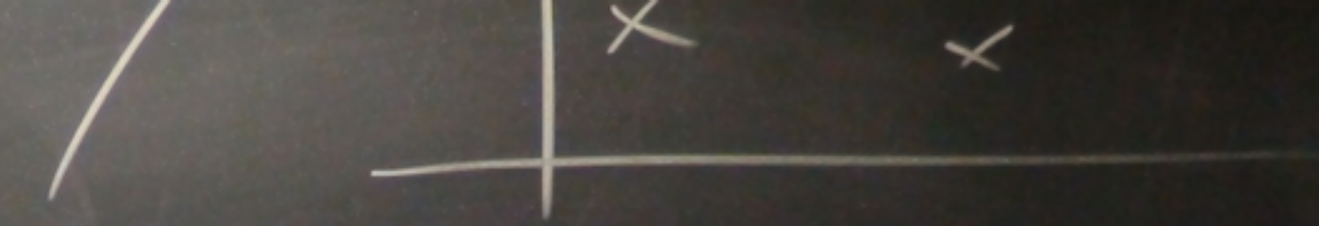
5.



(6) $\bar{x} = \frac{\sum_{n=1}^{N} x_n}{N}$

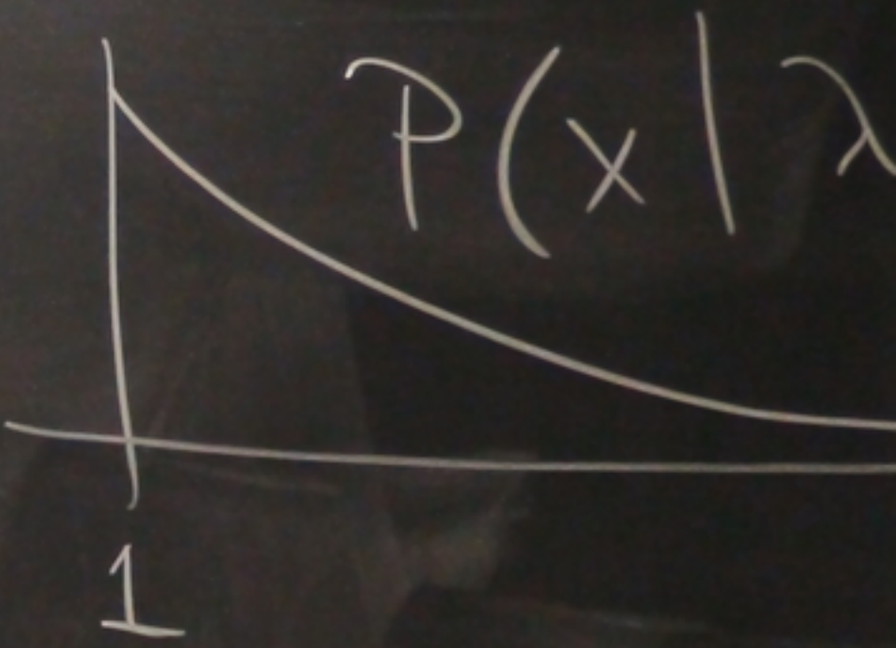
$\hat{\mu} = \bar{x}$

$\frac{\sum x_n}{N}$



@ Had to be chosen

$$\frac{\sum_{n=1}^{N}(x_n - a)}{N}$$

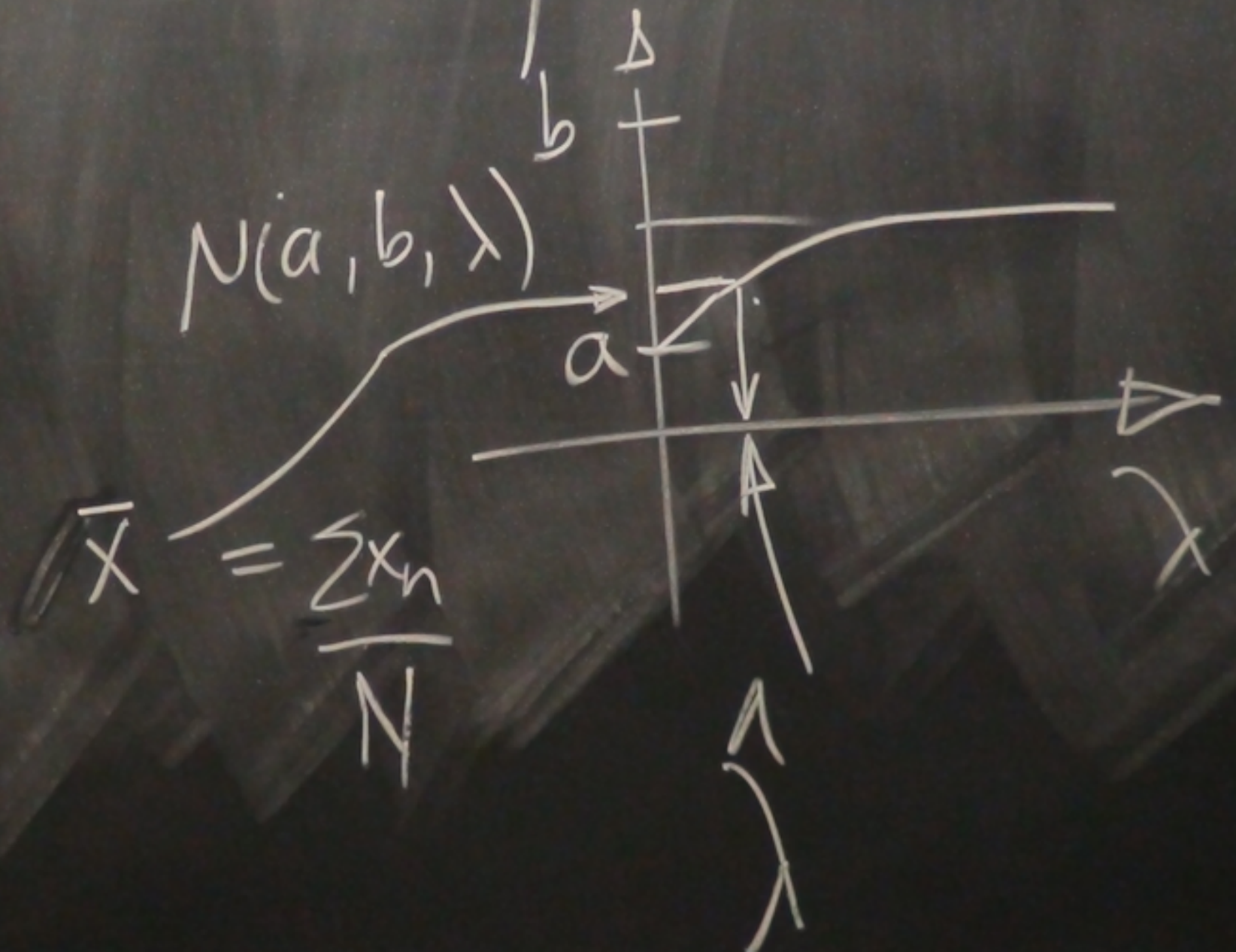


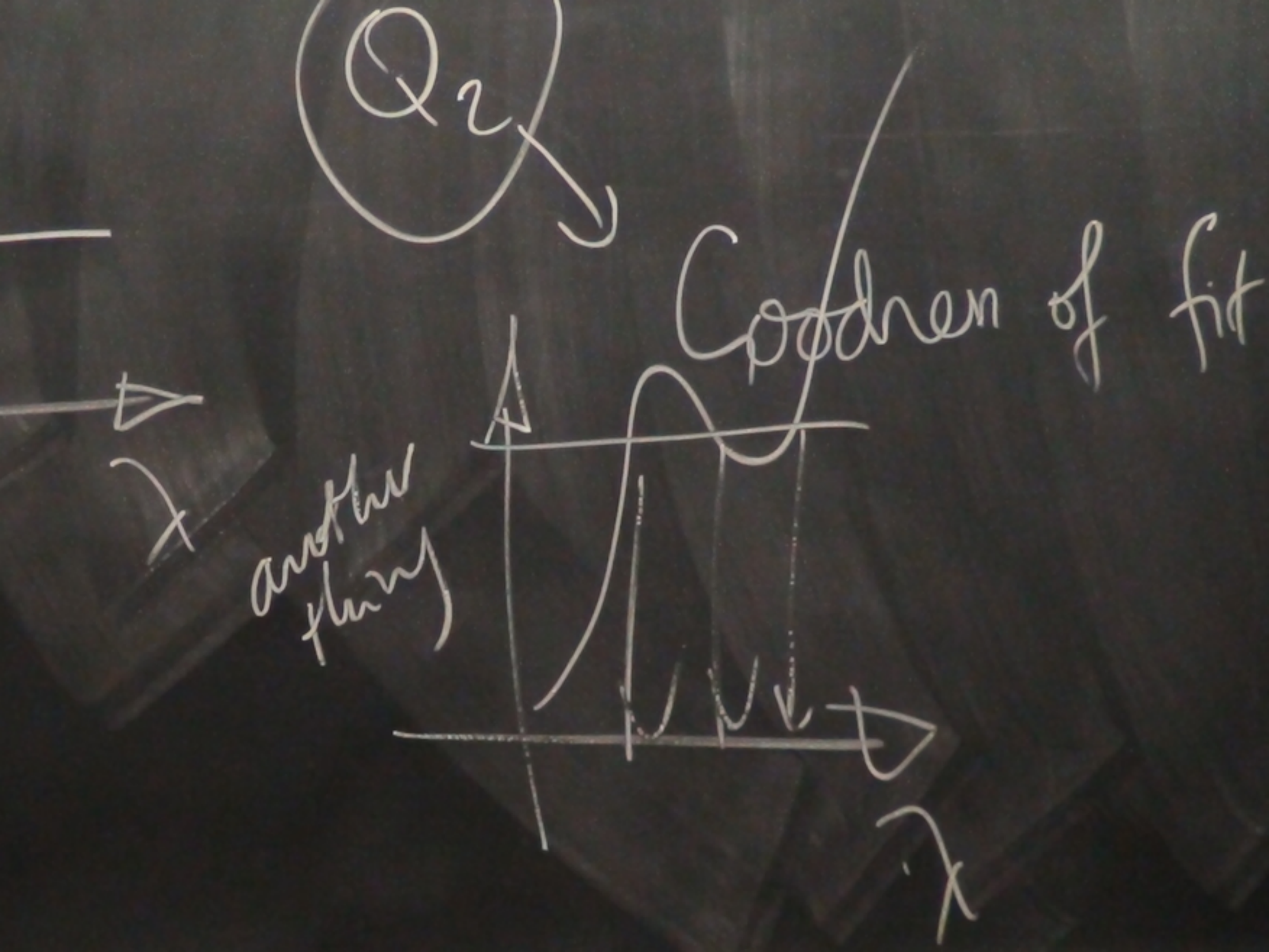
$$P(x|\lambda) = e^{-x/\lambda}/Z$$

$$\langle x \rangle = \lambda$$

$$\langle x \rangle = \frac{\int_a^b e^{-x/\lambda} \cdot x \, dx}{Z}$$

$$x \; dx = N(a, b, \lambda)$$




Q_2
Goodness of fit
another theory
χ^2
χ^2

Other ideas

- Chi-squared
  - minimize with respect to lambda
  - also depends on bins
- Other measure of 'error' between theory and data
- Find how the data-mean is expected to vary with lambda
- Other 'estimators', depending on other data-properties

The answer using Bayes's theorem

$$P(\lambda \,|\, \{x_n\}_{n=1}^N) = \frac{P(\{x_n\}_{n=1}^N \,|\, \lambda)\, P(\lambda)}{P(\{x_n\}_{n=1}^N)}$$

Or, making explicit the underlying assumptions -

$$P(\lambda \,|\, \{x_n\}_{n=1}^N, \mathcal{H}_1) = \frac{P(\{x_n\}_{n=1}^N \,|\, \lambda, \mathcal{H}_1)\, P(\lambda \,|\, \mathcal{H}_1)}{P(\{x_n\}_{n=1}^N \,|\, \mathcal{H}_1)}$$

$$\underbrace{P(\text{Parameters} \,|\, \text{Data}, \mathcal{H}_1)}_{\text{POSTERIOR}} = \frac{\overbrace{P(\text{Data} \,|\, \text{Parameters}, \mathcal{H}_1)}^{\text{LIKELIHOOD}}\ \overbrace{P(\text{Parameters} \,|\, \mathcal{H}_1)}^{\text{PRIOR}}}{\underbrace{P(\text{Data} \,|\, \mathcal{H}_1)}_{\text{EVIDENCE FOR } \mathcal{H}_1}}$$

$$p(x_n \mid \lambda, H) =$$

$$= \frac{e^{-x_n/\lambda}}{Z} \quad *$$

$$Z = \int_a^b e^{-x/\lambda} \, dx$$

$$= \lambda \left[e^{-a/\lambda} - e^{-b/\lambda} \right]$$

$$P(\lambda \mid \xi \times \xi, H) = \prod_{1}^{N} P(\xi \times \xi \mid \lambda, H)$$

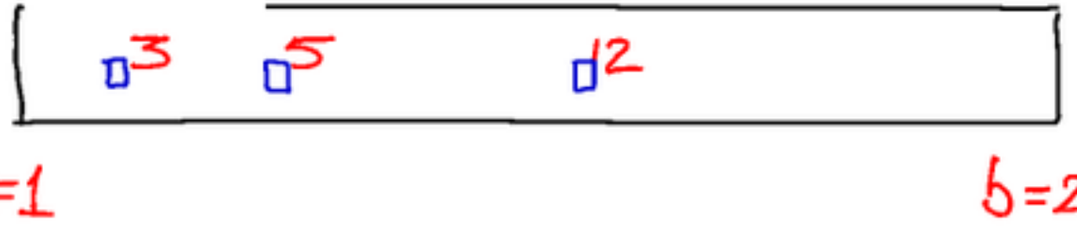
$$P(x \mid \lambda, H) = \prod_{1}^{N} \frac{1}{Z(\lambda, a, b)} e^{-\frac{x_n}{\lambda}}$$

$$P(\{x\}|\lambda,H) = \prod_{1}^{N} \frac{1}{Z(\lambda,a,b)} e^{-\frac{x_n}{\lambda}} \times P(\lambda|H)$$

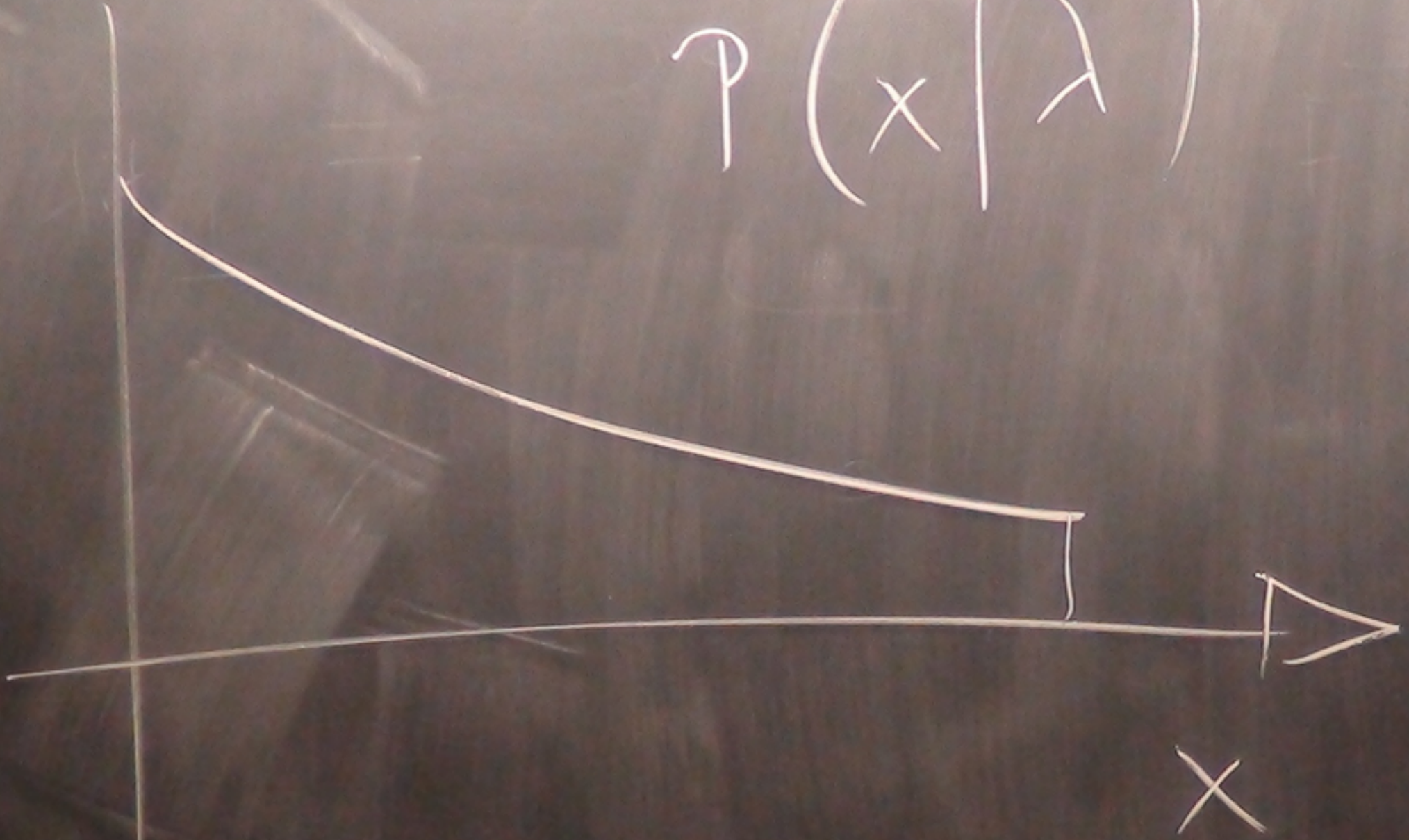
$$\frac{}{P(\{x\}|H)}$$

The answer using Bayes's theorem

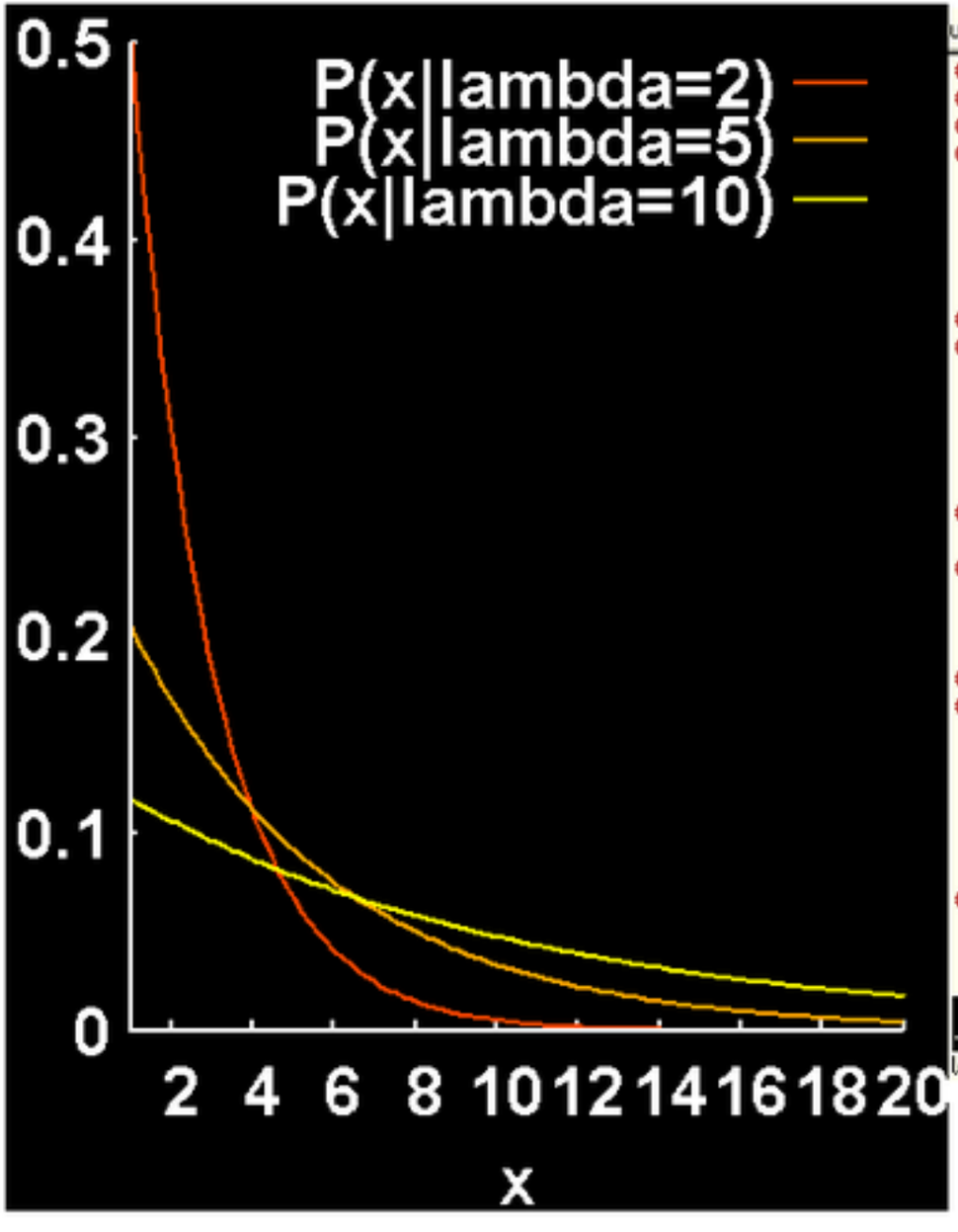
```
Buffers Files Tools Edit Search Mule Help
#
# gnuplot commands to illustrate how simple Bayesian inference is
#
# Define the density P(x|lambda)
        p(x,l) = exp ( -x/l ) / Z(l)/l
        Z(l) = exp ( -a/l) - exp(-b/l)
        a=1
        b=20

# The first plot illustrates the predictive
# distributions for different values of lambda
  plot [1:20][0:0.25] p(x,2) t "P(x|lambda=2)", \
                      p(x,5) t "P(x|lambda=5)", \
                      p(x,10) t "P(x|lambda=10)"


# Now imagine we receive data { x } = {3, 5, 12}

# This is the definition of the likelihood function

 like(l) =  p(3,l) *  p(5,l) *  p(12,l)

# The next plot shows the three factors, P(x_n|lambda),
# in the likelihood function
 set logscale x
 set xlabel "lambda"
 plot [0.3:140] p(3,x) t "P(x=3|lambda)", \
                p(5,x) t "P(x=5|lambda)", \
                p(12,x) t "P(x=12|lambda)"

# Finally, plot the likelihood

 plot [0.3:140] like(x) t 'P({x3}|lambda)' w l 5

--:--  gnu4b              (Perl)--L35--All----------------------------
Wrote /home/mackay/itp/octave/decay/gnu4b
```



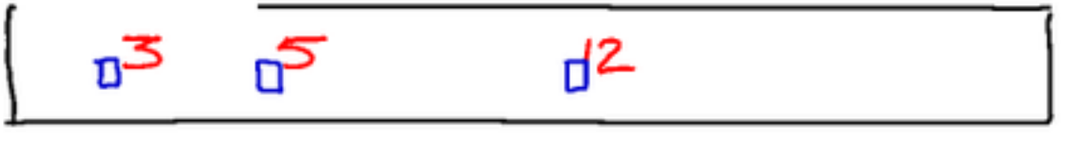
$P(x|\lambda)$



The answer using Bayes's theorem

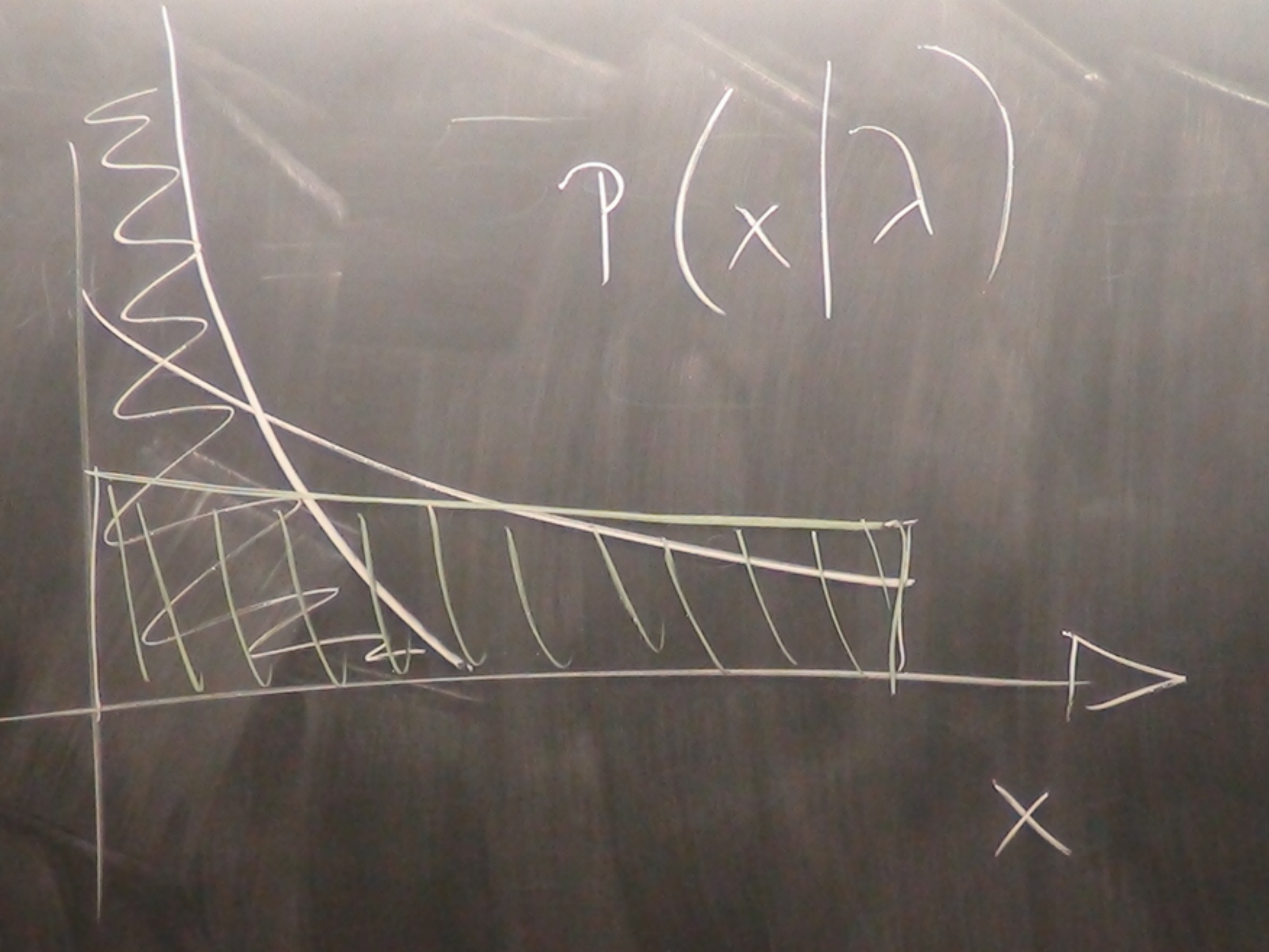



```
uffers Files Tools Edit Search Mule Help
#
# gnuplot commands to illustrate how simple Bayesian inference is
#
# Define the density P(x|lambda)
        p(x,l) = exp ( -x/l ) / Z(l)/l
        Z(l) = exp ( -a/l) - exp(-b/l)
        a=1
        b=20

# The first plot illustrates the predictive
# distributions for different values of lambda
  plot [1:20][0:0.25] p(x,2) t "P(x|lambda=2)", \
                      p(x,5) t "P(x|lambda=5)", \
                      p(x,10) t "P(x|lambda=10)"


# Now imagine we receive data { x } = {3, 5, 12}

# This is the definition of the likelihood function

 like(l) =  p(3,l) *  p(5,l) *  p(12,l)

# The next plot shows the three factors, P(x_n|lambda),
# in the likelihood function
 set logscale x
 set xlabel "lambda"
 plot [0.3:140] p(3,x) t "P(x=3|lambda)", \
                p(5,x) t "P(x=5|lambda)", \
                p(12,x) t "P(x=12|lambda)"

# Finally, plot the likelihood

 plot [0.3:140] like(x) t 'P({x3}|lambda)' w l 5

--:--  gnu4b            (Perl)--L35--All----------------------------
Wrote /home/mackay/itp/octave/decay/gnu4b
```

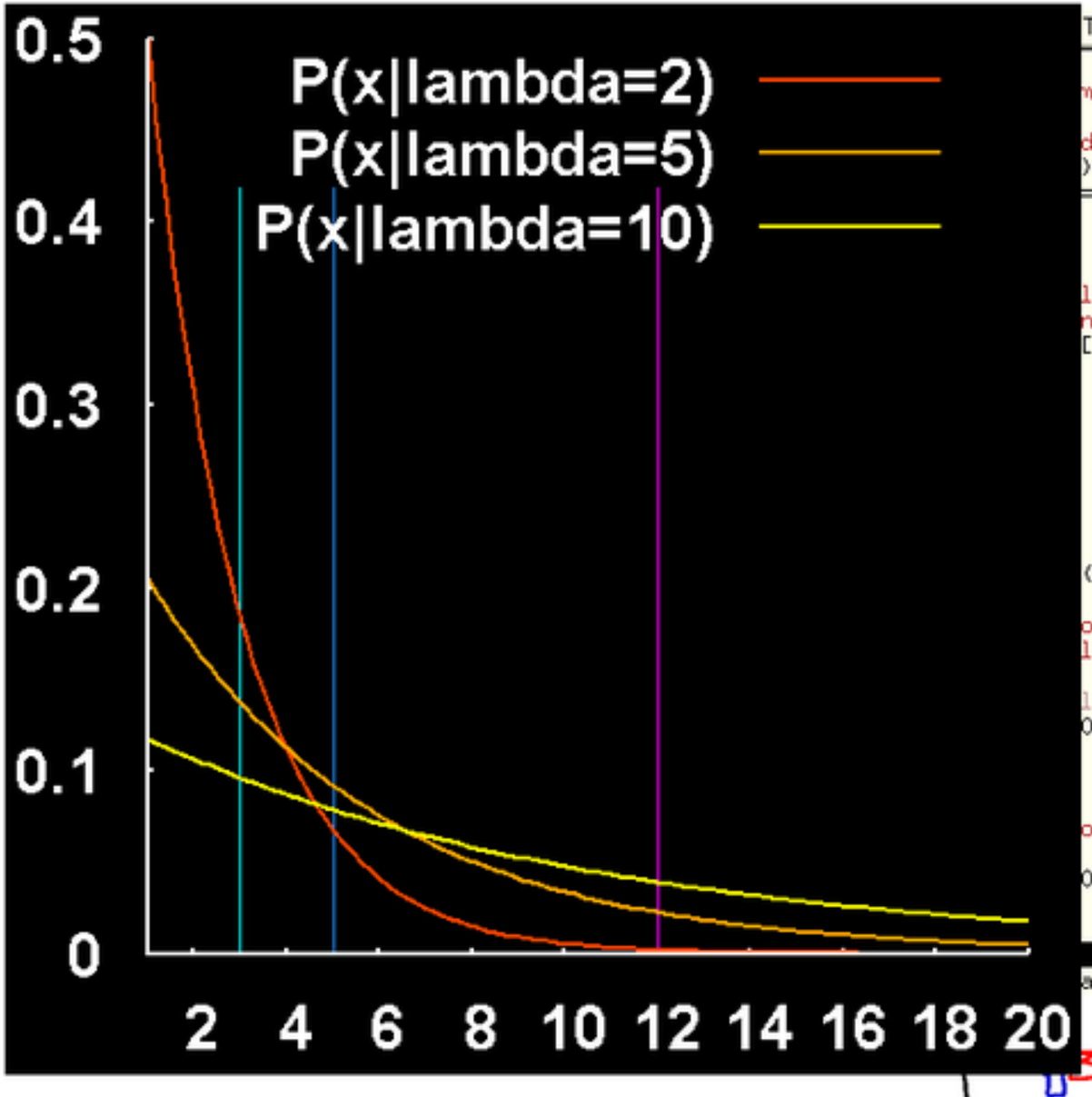


a=1

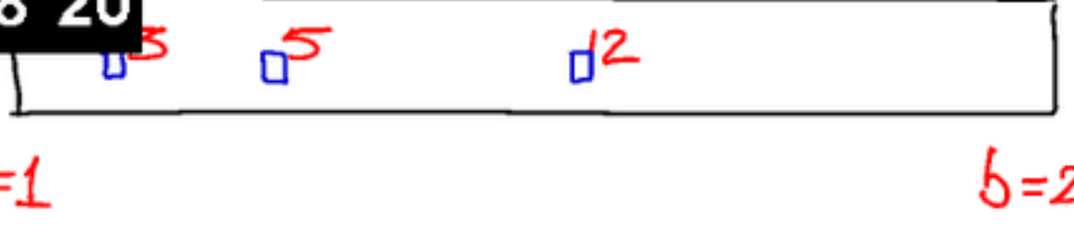
b=20


$p(x|\lambda)$
x

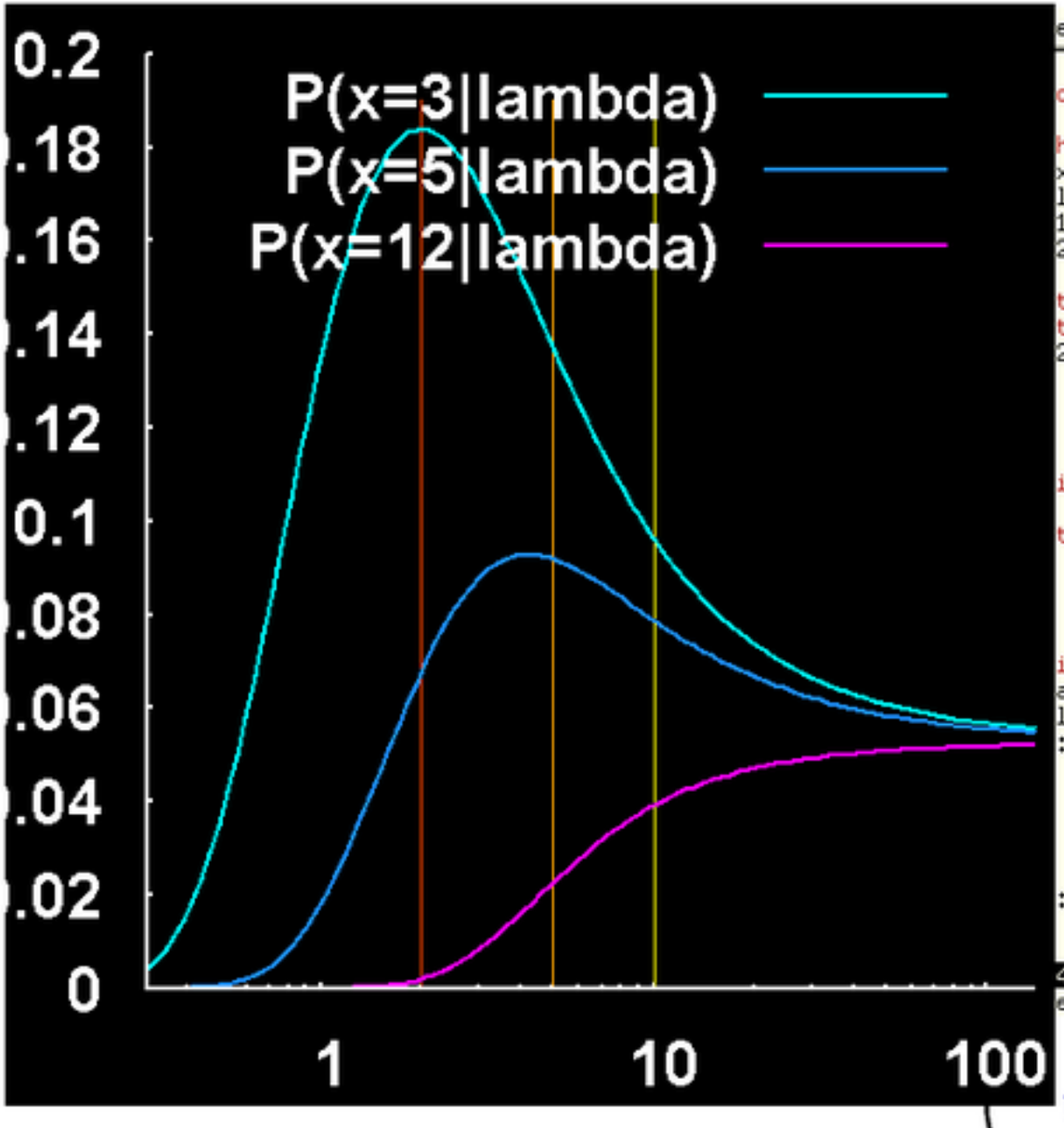
The answer using Bayes's theorem



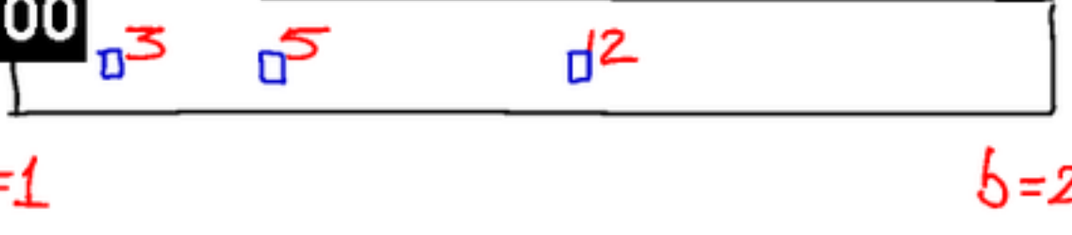

```
Tools Edit Search Mule Help

mands to illustrate how simple Bayesian inference is

density P(x|lambda)
) = exp ( -x/l ) / Z(l)/l
= exp ( -a/l) - exp(-b/l)



lot illustrates the predictive
ns for different values of lambda
[0:0.25] p(x,2) t "P(x|lambda=2)", \
         p(x,5) t "P(x|lambda=5)", \
         p(x,10) t "P(x|lambda=10)"


 we receive data { x } = {3, 5, 12}

 definition of the likelihood function

(3,l) *  p(5,l) *  p(12,l)

ot shows the three factors, P(x_n|lambda),
lihood function
 x
lambda"
0] p(3,x) t "P(x=3|lambda)", \
   p(5,x) t "P(x=5|lambda)", \
   p(12,x) t "P(x=12|lambda)"

ot the likelihood

0] like(x) t 'P({x3}|lambda)' w l 5


                (Perl)--L35--All----------------------------
ackay/itp/octave/decay/gnu4b
```

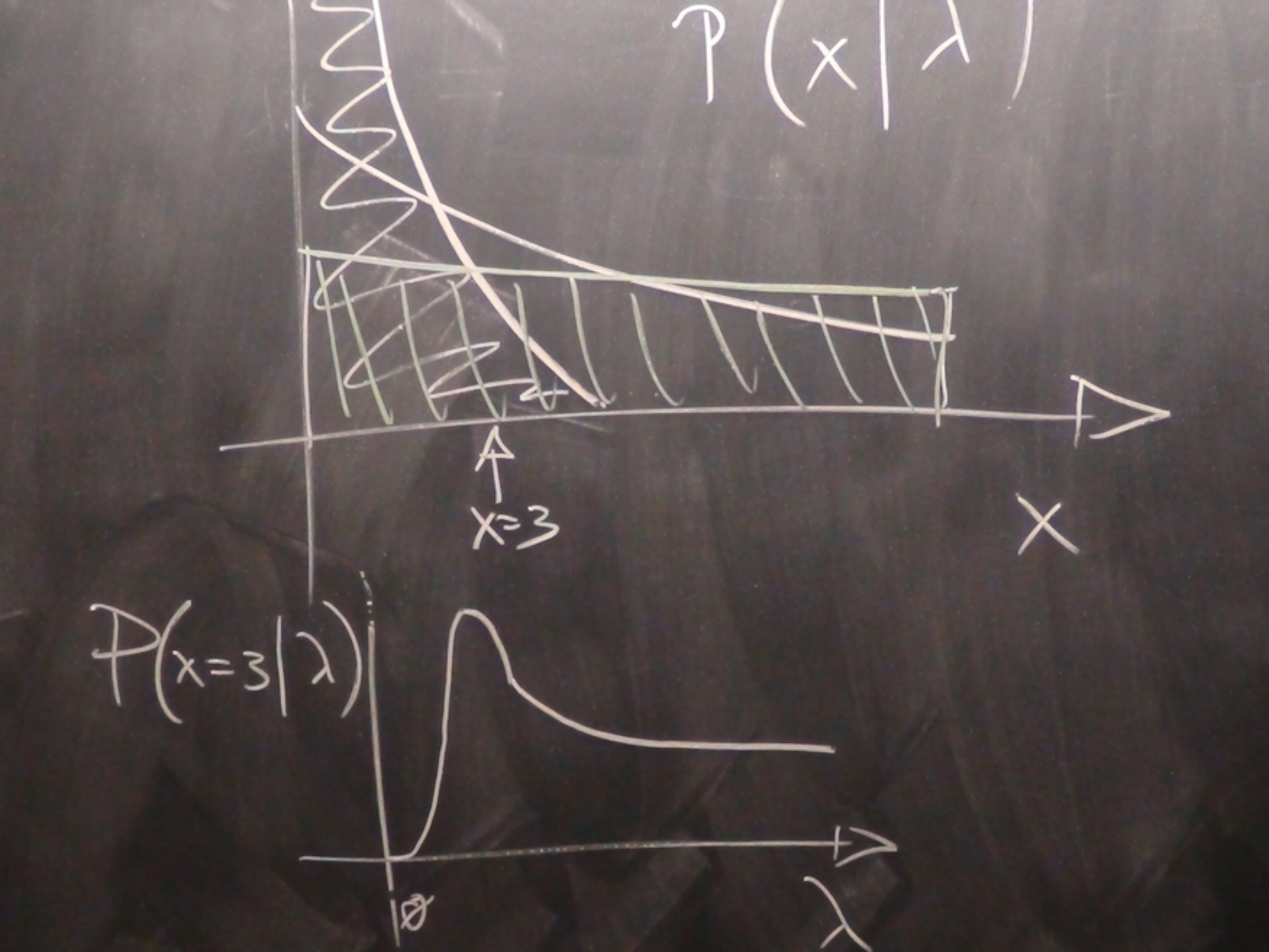


The answer using Bayes's theorem

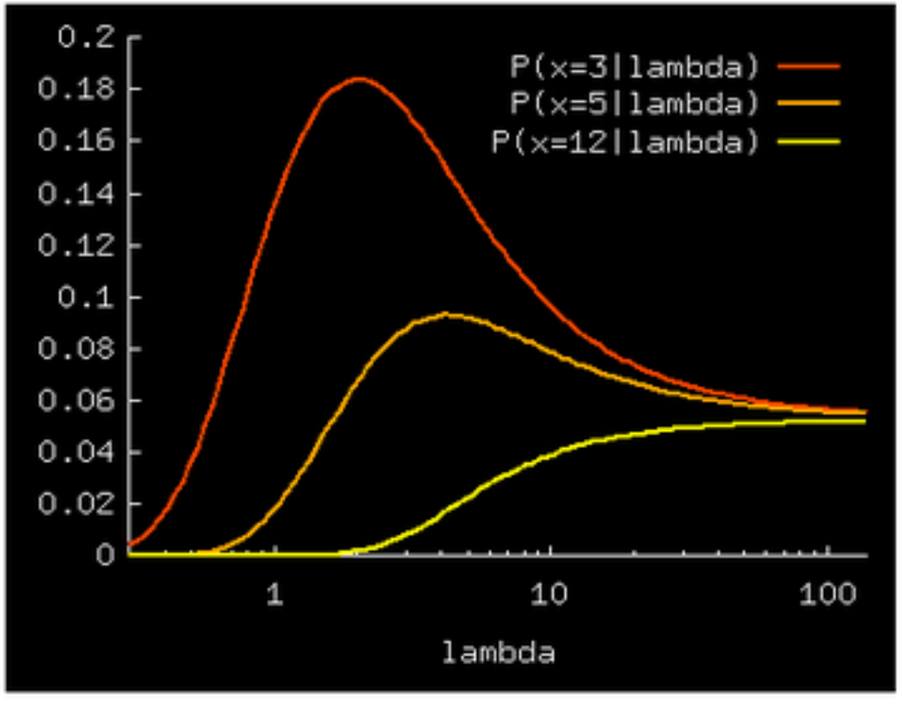


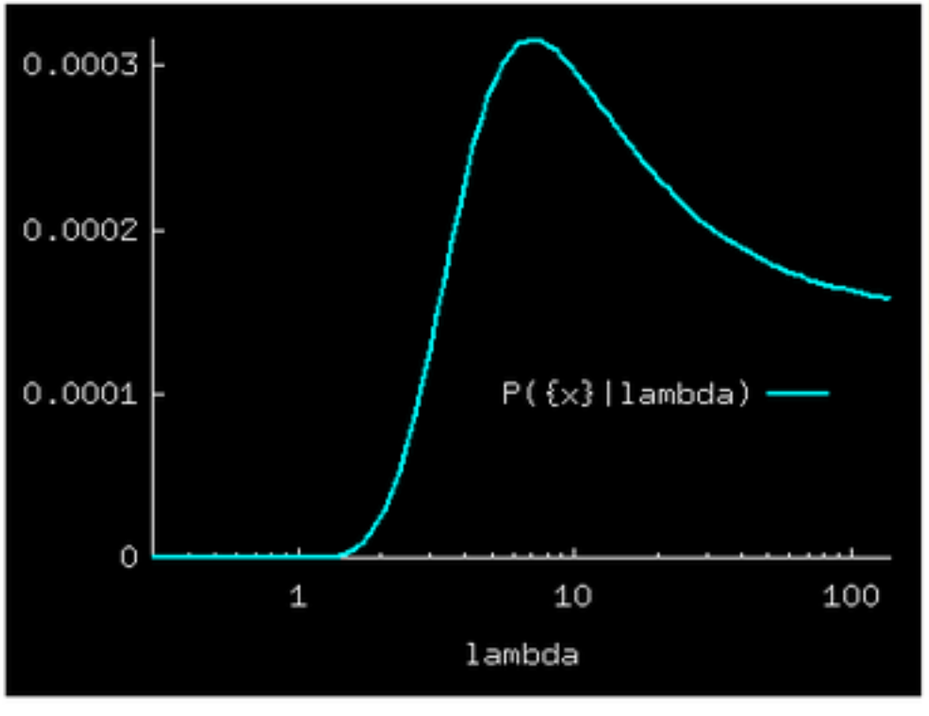

```
es Tools Edit Search Mule Help

commands to illustrate how simple Bayesian inference is

he density P(x|lambda)
x,l) = exp ( -x/l ) / Z(l)/l
l) = exp ( -a/l) - exp(-b/l)
1
20

t plot illustrates the predictive
tions for different values of lambda
20][0:0.25] p(x,2) t "P(x|lambda=2)", \
            p(x,5) t "P(x|lambda=5)", \
            p(x,10) t "P(x|lambda=10)"


ine we receive data { x } = {3, 5, 12}

the definition of the likelihood function

  p(3,l) *  p(5,l) *  p(12,l)

 plot shows the three factors, P(x_n|lambda),
ikelihood function
ale x
l "lambda"
:140] p(3,x) t "P(x=3|lambda)", \
      p(5,x) t "P(x=5|lambda)", \
      p(12,x) t "P(x=12|lambda)"

 plot the likelihood

:140] like(x) t 'P({x3}|lambda)' w l 5


4b            (Perl)--L35--All----------------------------
e/mackay/itp/octave/decay/gnu4b
```



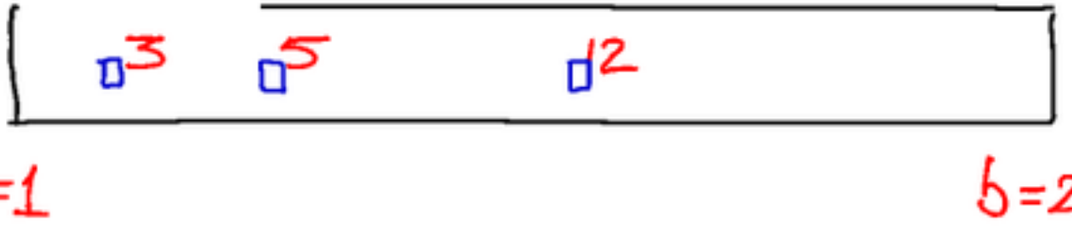

P(x|λ)
x=3
X
P(x=3|λ)
0
λ

The answer using Bayes's theorem



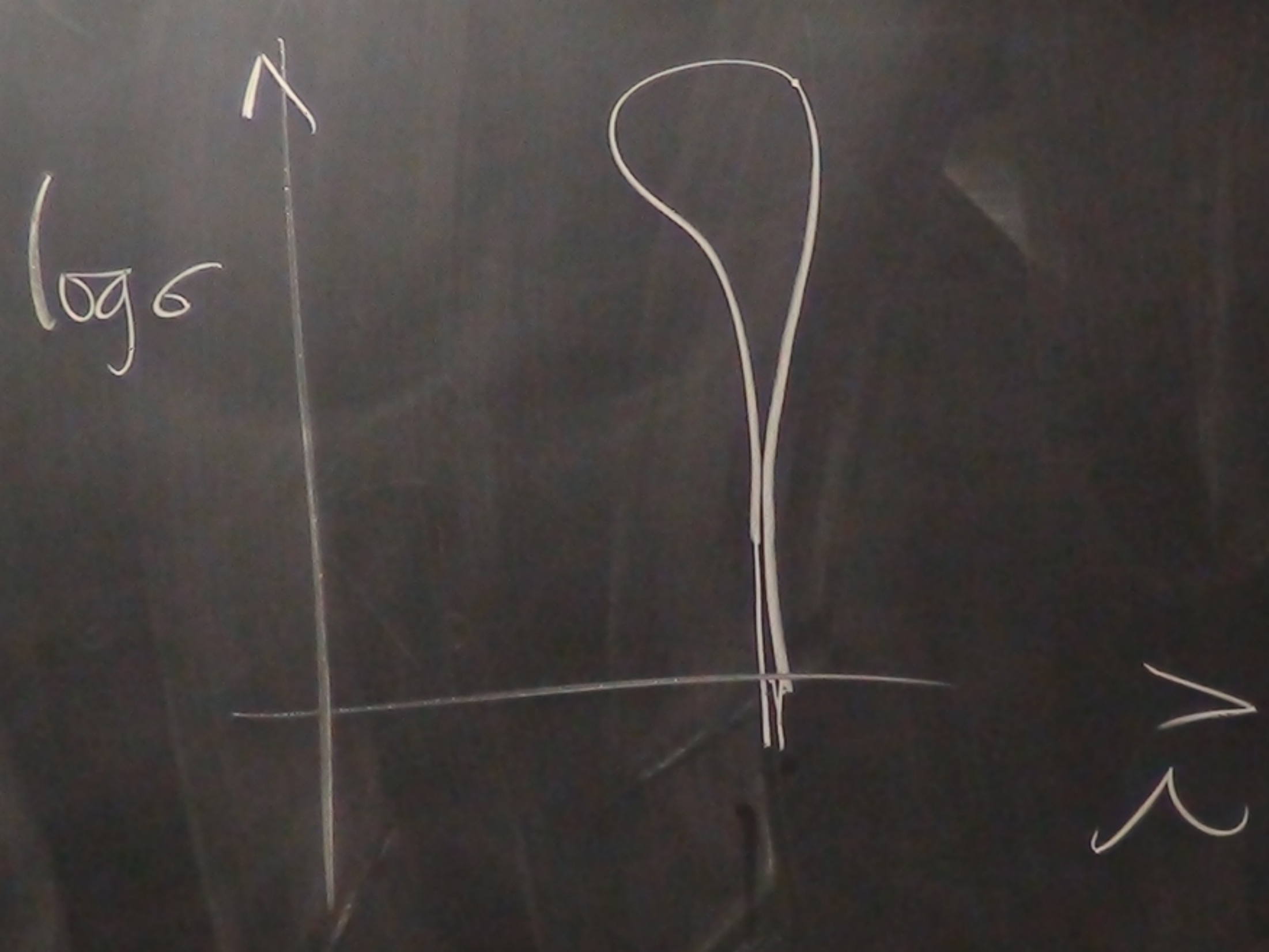



```
Buffers Files Tools Edit Search Mule Help
#
# gnuplot commands to illustrate how simple Bayesian inference is
#
# Define the density P(x|lambda)
        p(x,l) = exp ( -x/l ) / Z(l)/l
        Z(l) = exp ( -a/l) - exp(-b/l)
        a=1
        b=20

# The first plot illustrates the predictive
# distributions for different values of lambda
  plot [1:20][0:0.25] p(x,2) t "P(x|lambda=2)", \
                      p(x,5) t "P(x|lambda=5)", \
                      p(x,10) t "P(x|lambda=10)"


# Now imagine we receive data { x } = {3, 5, 12}

# This is the definition of the likelihood function

 like(l) =  p(3,l) *  p(5,l) *  p(12,l)

# The next plot shows the three factors, P(x_n|lambda),
# in the likelihood function
 set logscale x
 set xlabel "lambda"
 plot [0.3:140] p(3,x) t "P(x=3|lambda)", \
                p(5,x) t "P(x=5|lambda)", \
                p(12,x) t "P(x=12|lambda)"

# Finally, plot the likelihood

 plot [0.3:140] like(x) t 'P({x}|lambda)' w l 5

--:--  gnu4b              (Perl)--L35--All----------------------------
Wrote /home/mackay/itp/octave/decay/gnu4b
```



$\xi \times \xi, H) = \prod \frac{1}{Z(\lambda, a, b)}$

$$\text{likelihood} = \frac{1}{Z(\lambda, a, b)^N} e^{-\frac{\sum x_n}{\lambda}}$$


log
x

$$\frac{1}{-}\frac{x_n}{\lambda}$$

$$e \qquad \times \; P(\lambda|H)$$

$$,b)$$

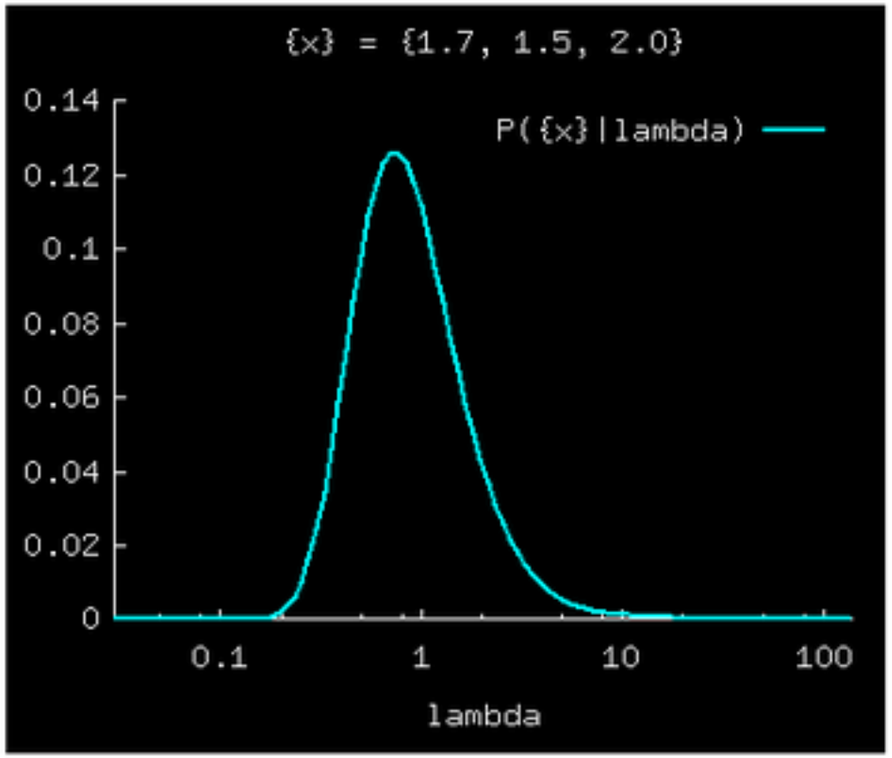
$$P(\xi \times \xi | H)$$

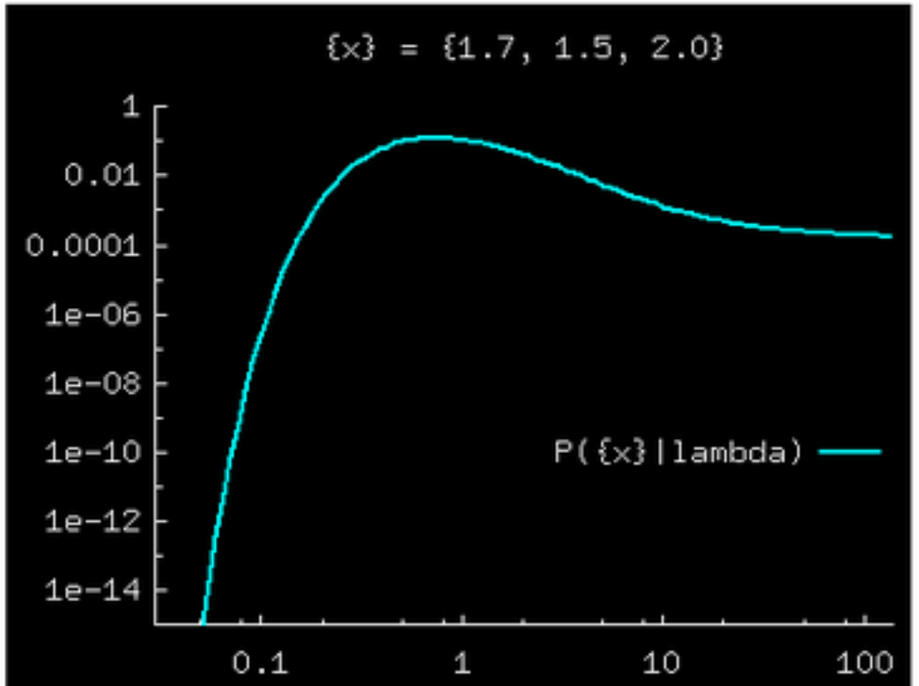
$$\underline{\underline{2}}$$

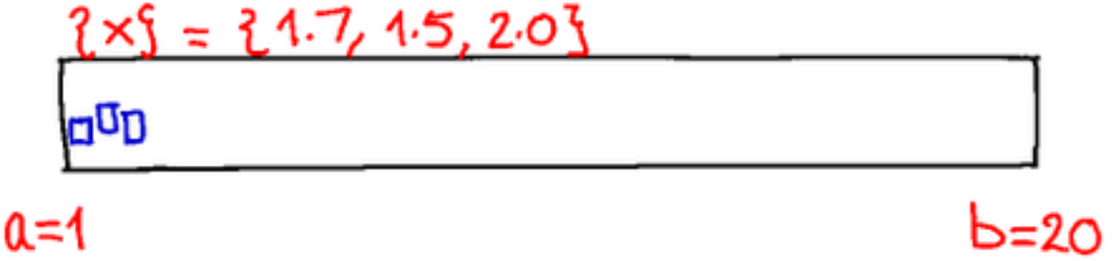
$$\rightarrow \text{approx likelihood by a Gaussian ?}$$

$$\overline{F}$$

Bayesian answer, alternate data set

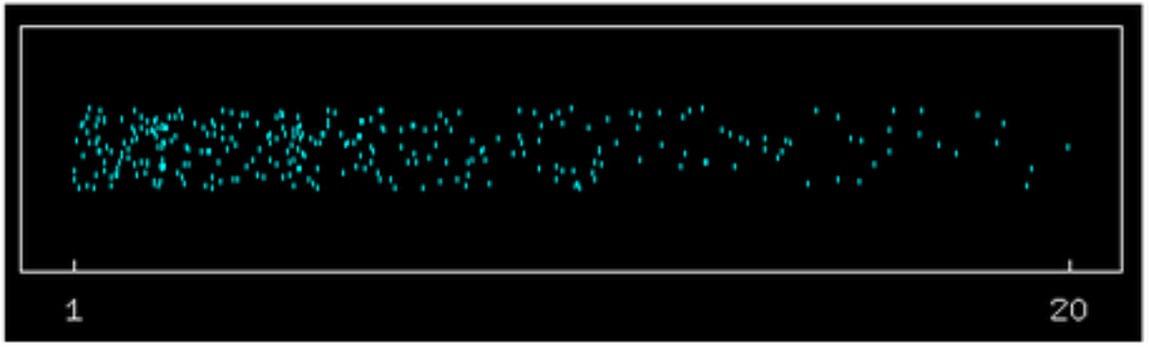







```
set title "{x} = {1.7, 1.5, 2.0}"
# This is the definition of the likelihood function

 like(l) =  p(1.7,l) *  p(1.5,l) *  p(2,l)
```

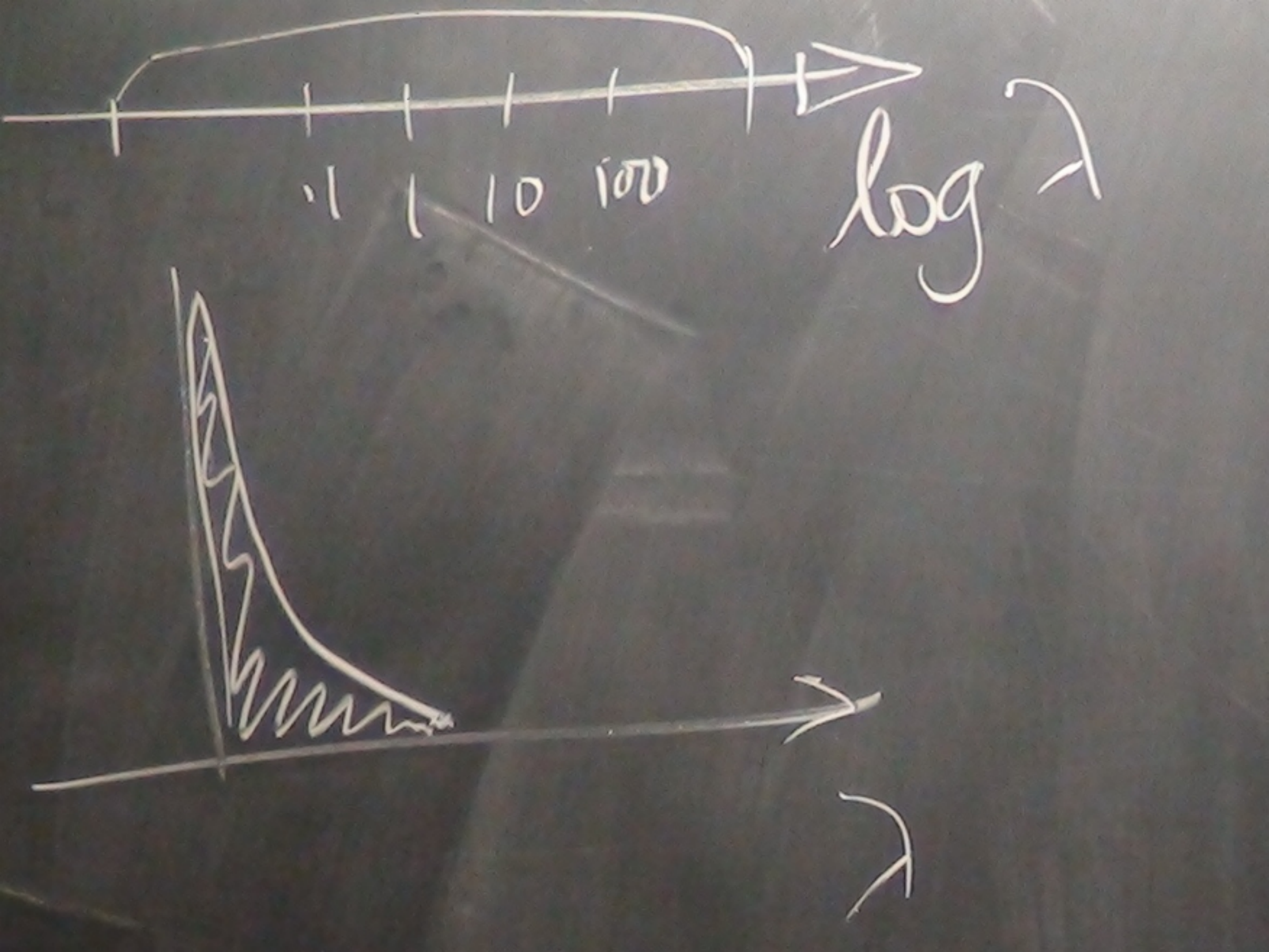
Inferring parameters in science experiments



A source emits particles; they decay at locations exponentially distributed with lengthscale λ. Decays can be observed only if the location falls in a window $(x_{\min}, x_{\max}) = (a, b) = (1, 20)$.

N decays are observed at locations $\{x_1, \ldots, x_N\}$. What is λ?

+ What about the possibility that the distribution is a mixture of two exponentials?


0.1
1
10
100
log λ
λ

$$P(x \mid \lambda_1, \lambda_2, \pi_1, \pi_2, H_2)$$

$a = 0$

$\pi_1 > 0$ $\boxed{\pi_1 = 1/2}$

$\pi_1 + \pi_2 = 1$

$$= \begin{cases} \left(\pi_1 \frac{e^{-x/\lambda_1}}{\lambda_1} + \pi_2 \frac{e^{-x/\lambda_2}}{\lambda_2} \right) & x \in (a, b) \\ 0 & \text{otherwise} \end{cases}$$

$$P(\lambda_1, \lambda_2 \mid \xi \times \xi, H) = \frac{P(\xi \times \xi \mid \lambda_1, \lambda_2, H_2)\, P(\lambda_1, \lambda_2 \mid H_2)}{P(\xi \times \xi \mid H_2)}$$

Model comparison

$$P(H_1 \mid \xi \times \xi, I) = \frac{P(\xi \times \xi \mid H_1, I)\, P(H_1 \mid I)}{P(\xi \times \xi \mid I)}$$

$$P(H_2 \mid \xi \times \xi, I) = \frac{P(\xi \times \xi \mid H_2, I)\, P(H_2 \mid I)}{P(\xi \times \xi \mid I)}$$

$$P(\xi \times \xi \mid \lambda_1, \lambda_2, H_2)\, P(\lambda_1, \lambda_2 \mid H_2)$$

$$P(\xi \times \xi \mid H_2)$$

$$= P(\xi \times \xi \mid H_1, I)\, P(H_1 \mid I)$$

$$= \frac{P(\xi \times \xi \mid H_2, I)\, P(H_2 \mid I)}{P(\xi \times \xi \mid I)}$$

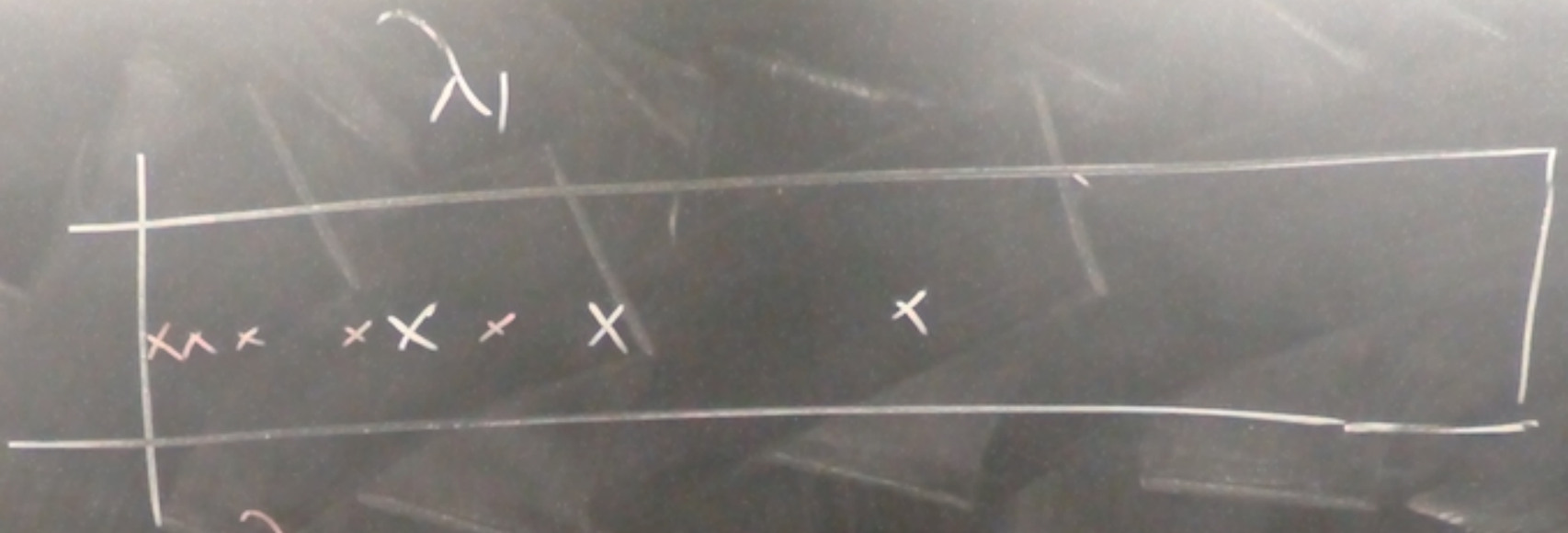
$$\frac{1}{Z(\lambda, a, b)} e$$

$$\times P(\lambda |$$

$$P(\xi \times \xi | H_1$$

$$e^{-\frac{\sum_1^N x_n}{\lambda}}$$

$$a, b)^N$$



$$P\left(\{x_n\}_{n=1}^{N}, \{k_n\}_{n=1}^{N} \middle| H_2, \lambda_1, \lambda_2\right)$$

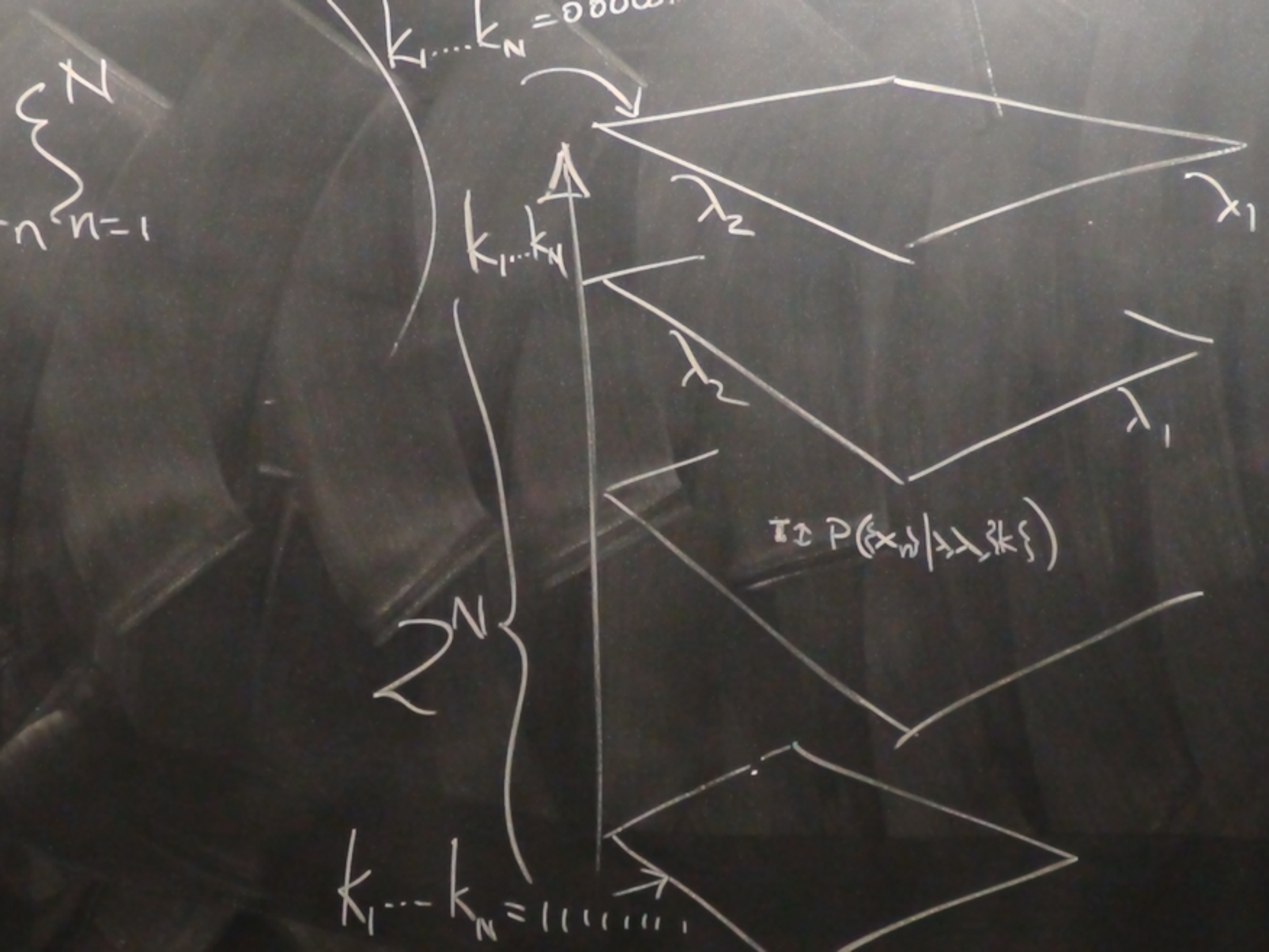
$$\lambda_2$$

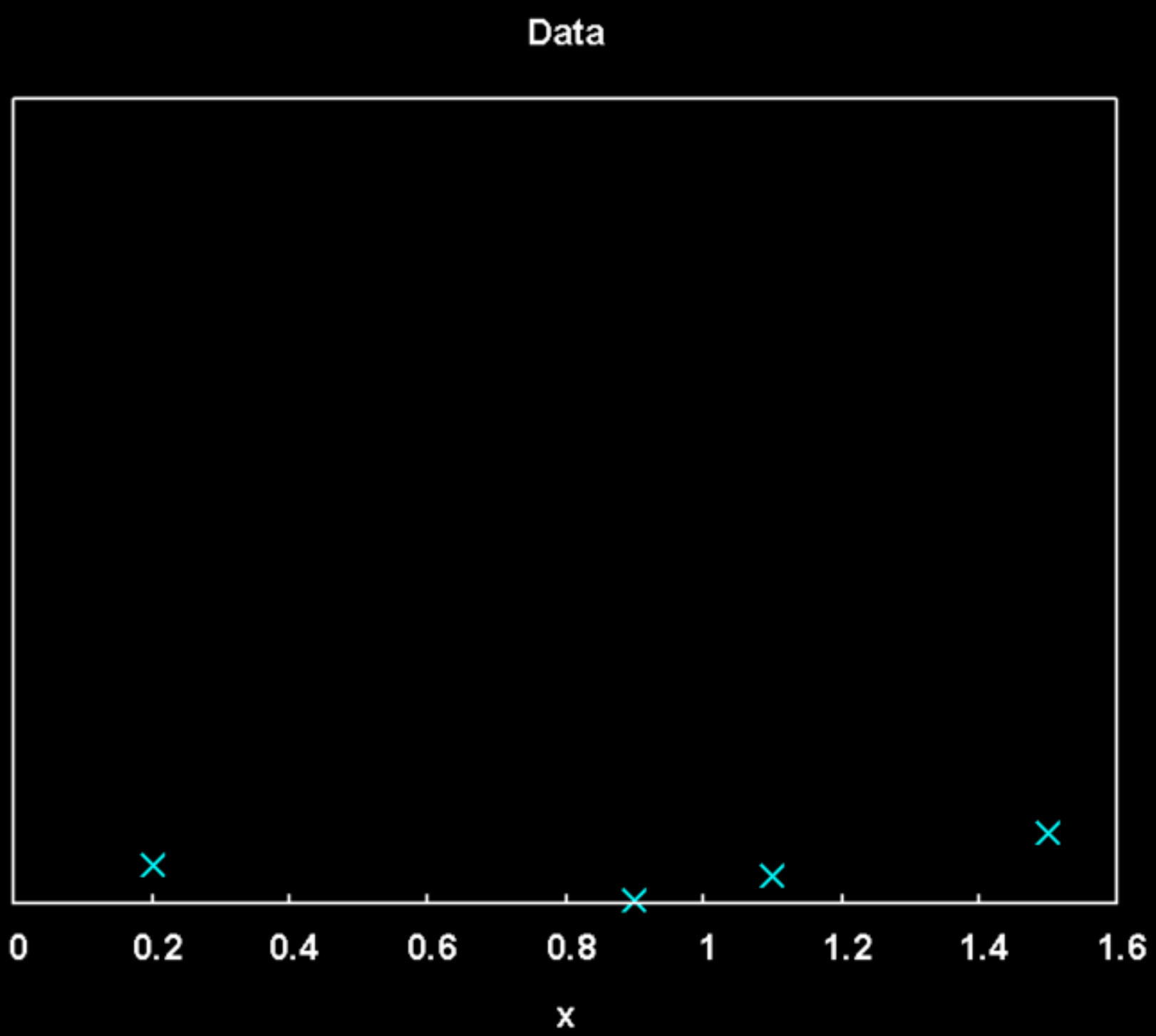
$$\{x_n\}_{n=1}^{N}, \{k_n\}_{n=1}^{N} \mid H_2, \lambda_1, \lambda_2$$

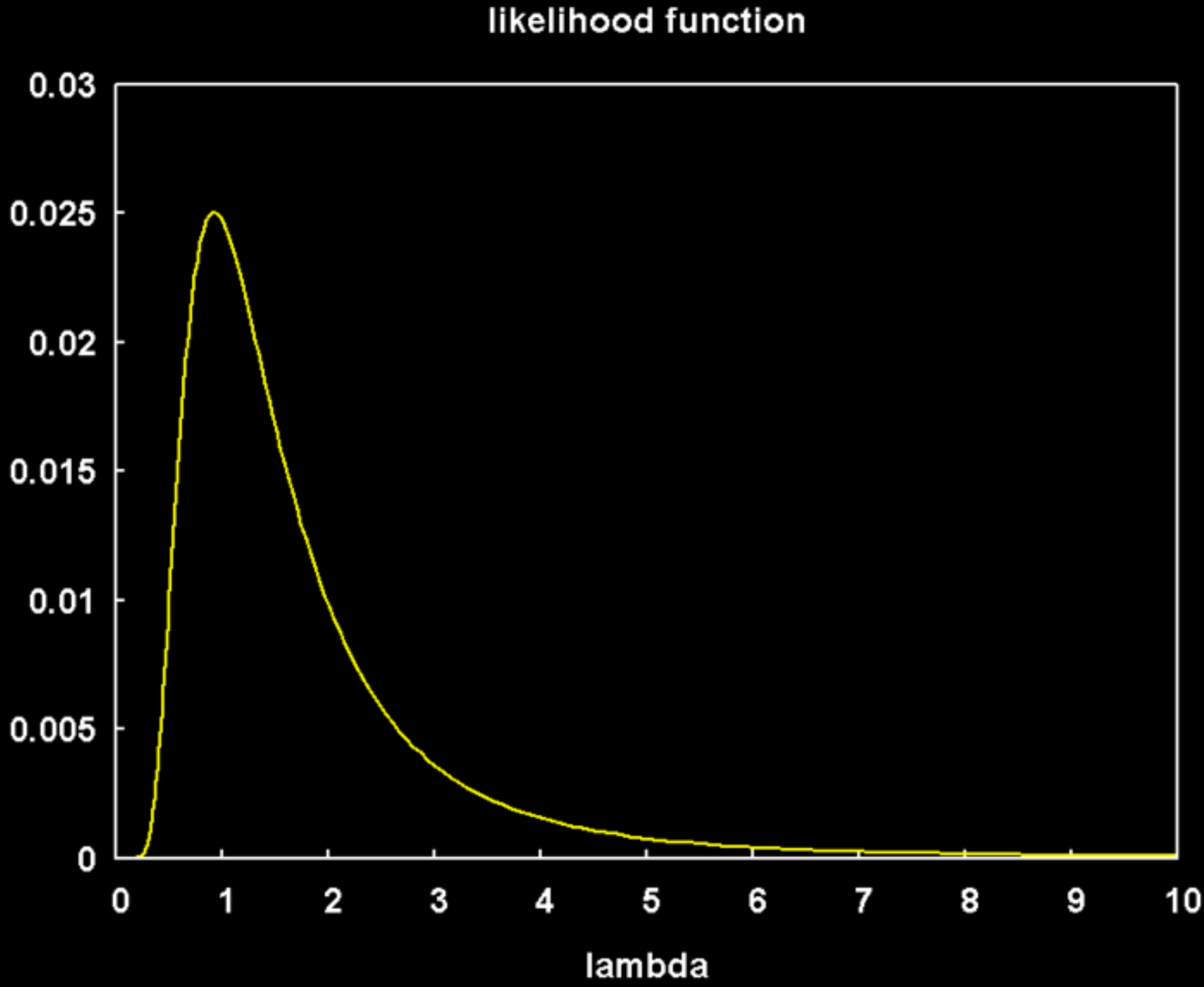
$$P(\{k_n\}_1^N \mid \{x_n\}_1^N, H_2)$$

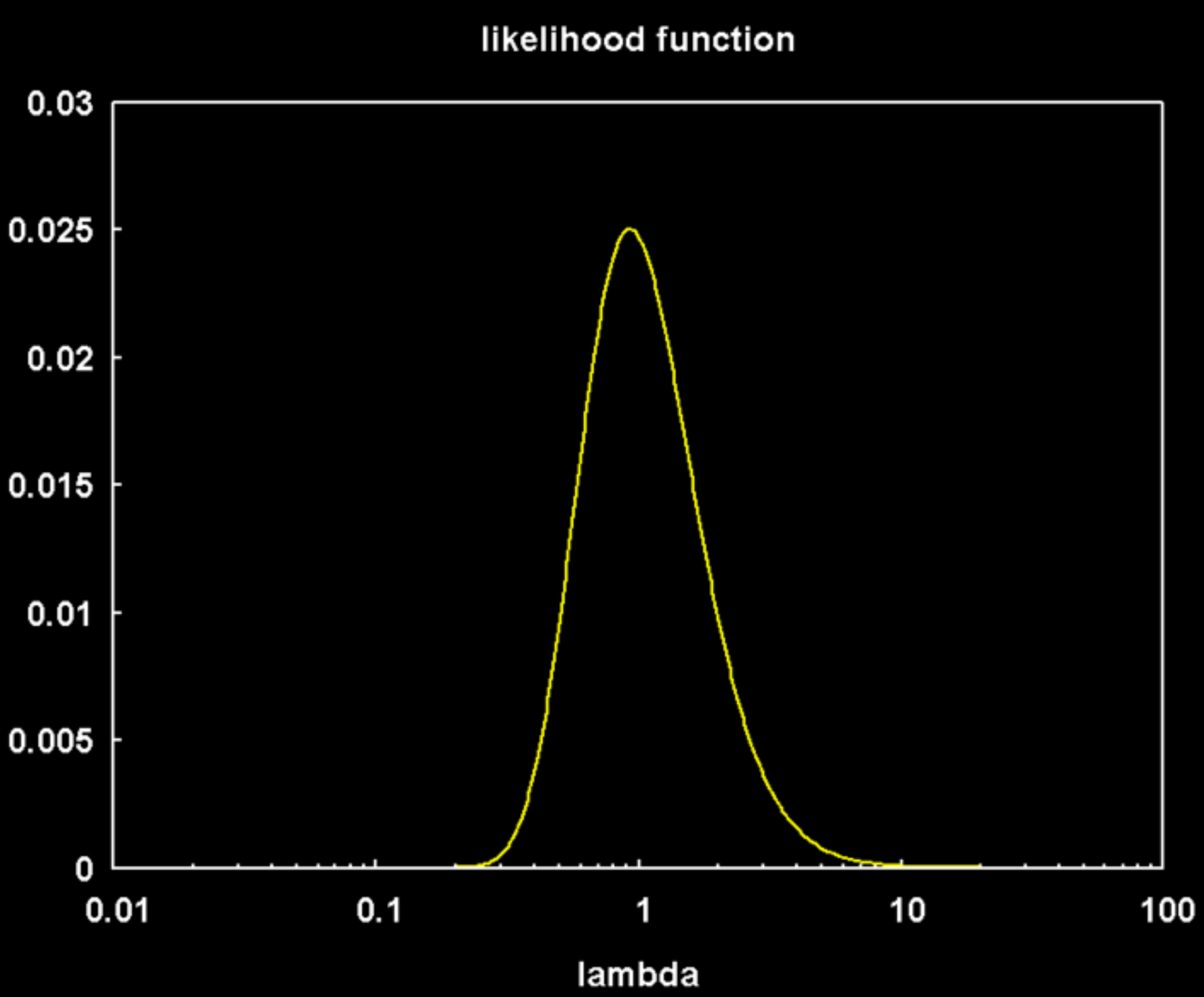
$$P(\lambda_1, \lambda_2 \mid \{x_n\}_1^N, H_2)$$

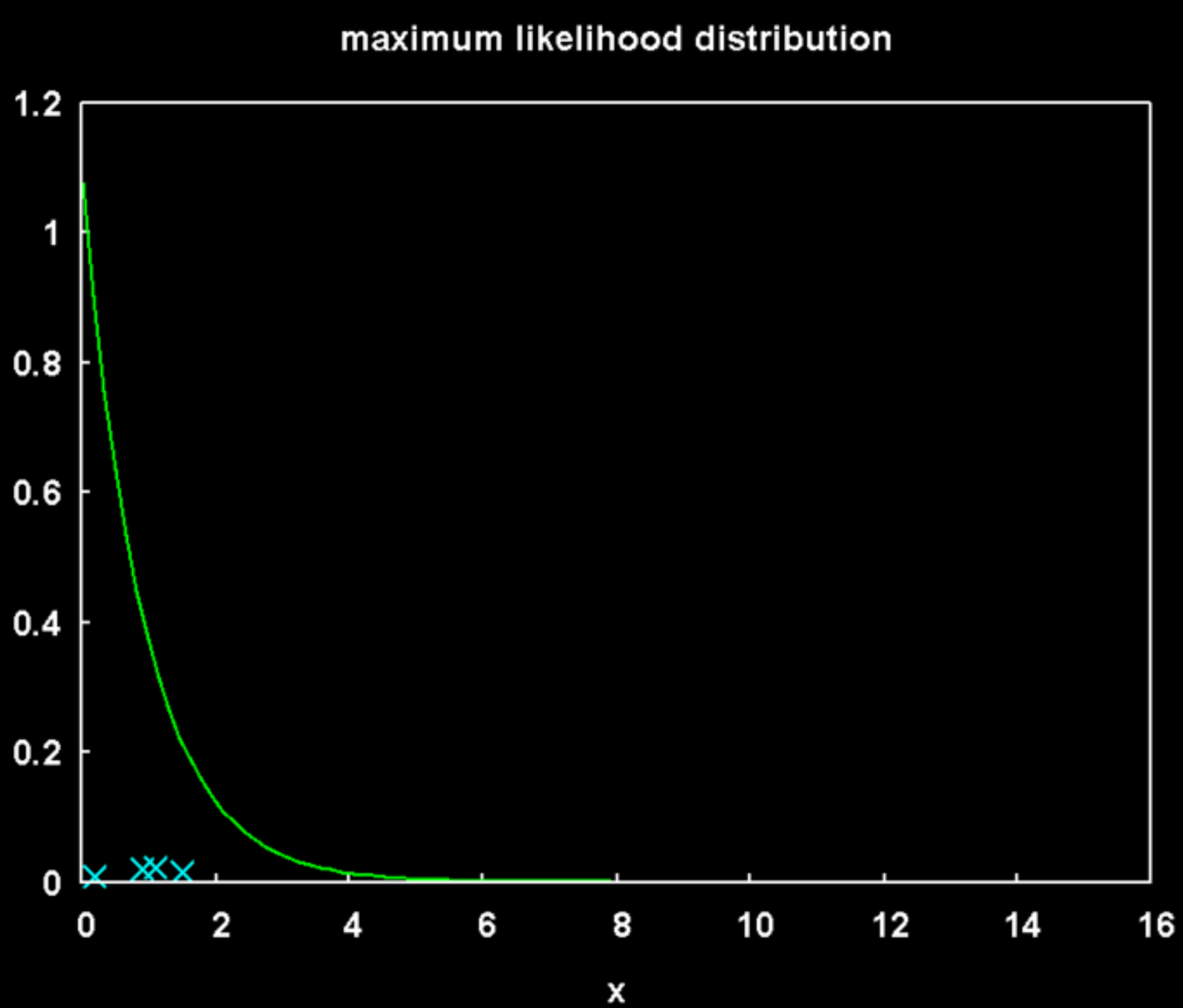
$$P(\xi \times \xi_1^N \mid \lambda_1, \lambda_2, \xi k_n \xi_{n=}^N$$

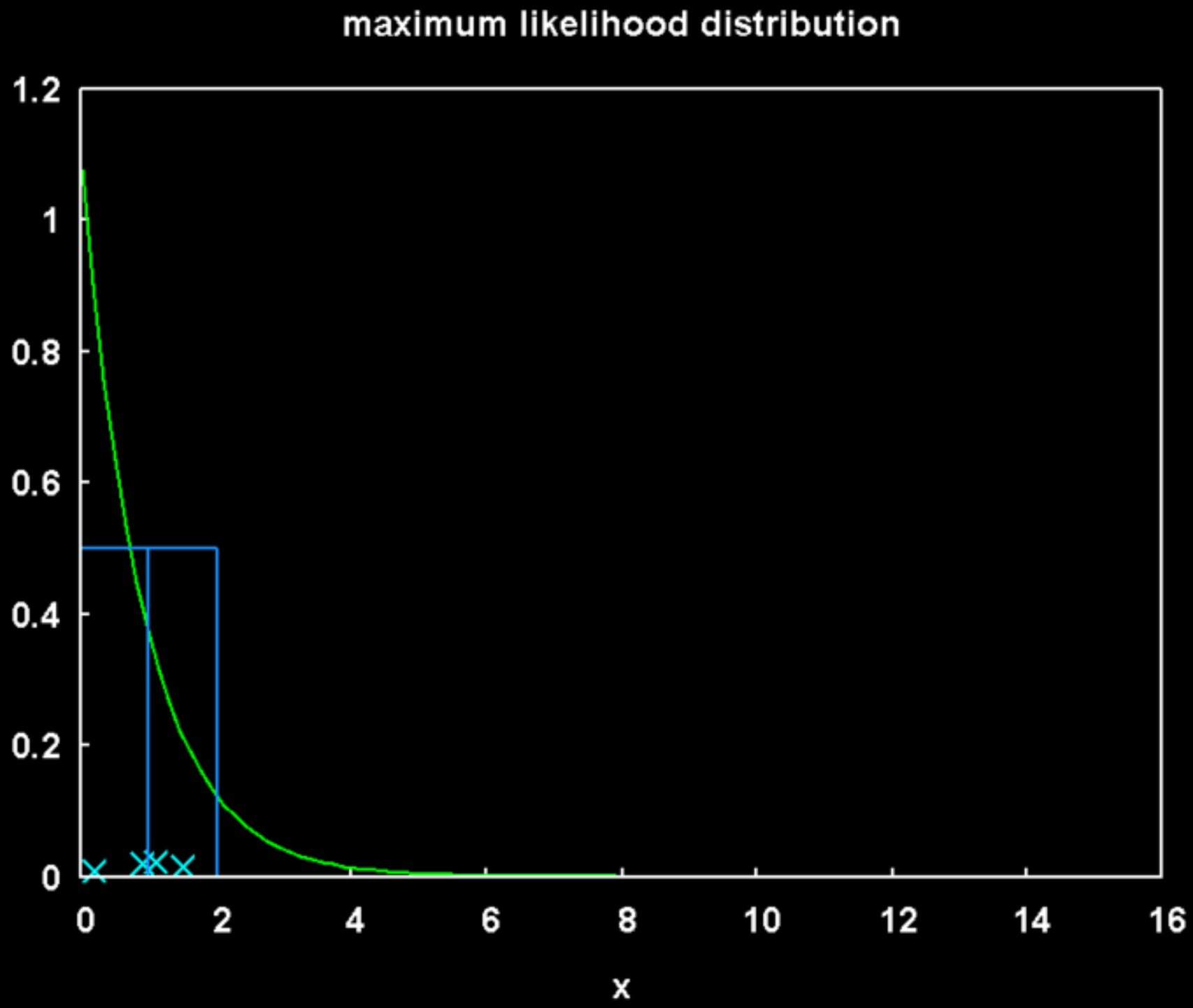

N
n=1
$k_1 \dots k_N$
λ_2
λ_1
$k_1 \dots k_N$
λ_2
λ_1
$\prod P(\{x_n\}|\lambda_1\lambda_2\{k\})$
2N
$k_1 \dots k_N = 1111111$

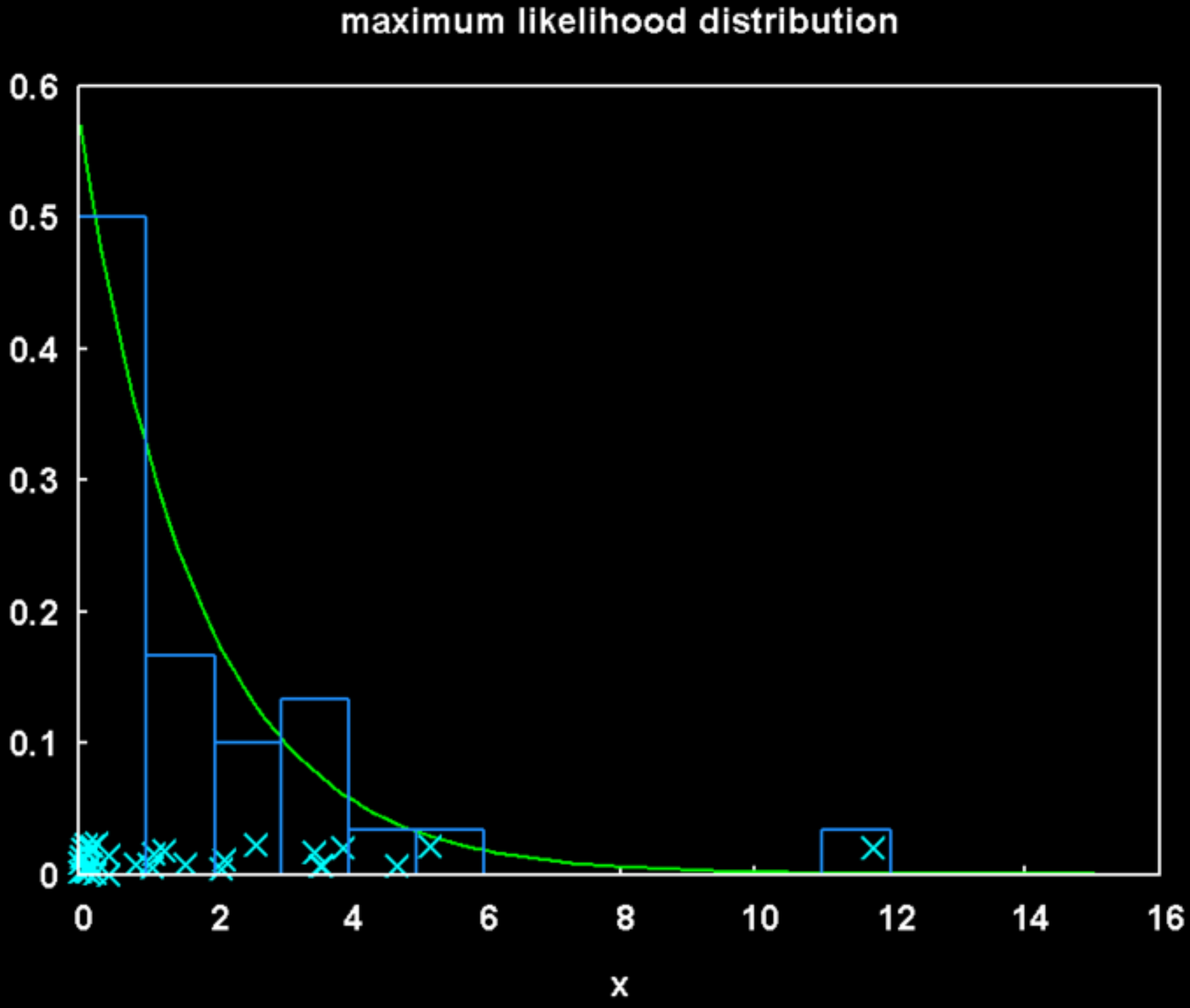

Data
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
x

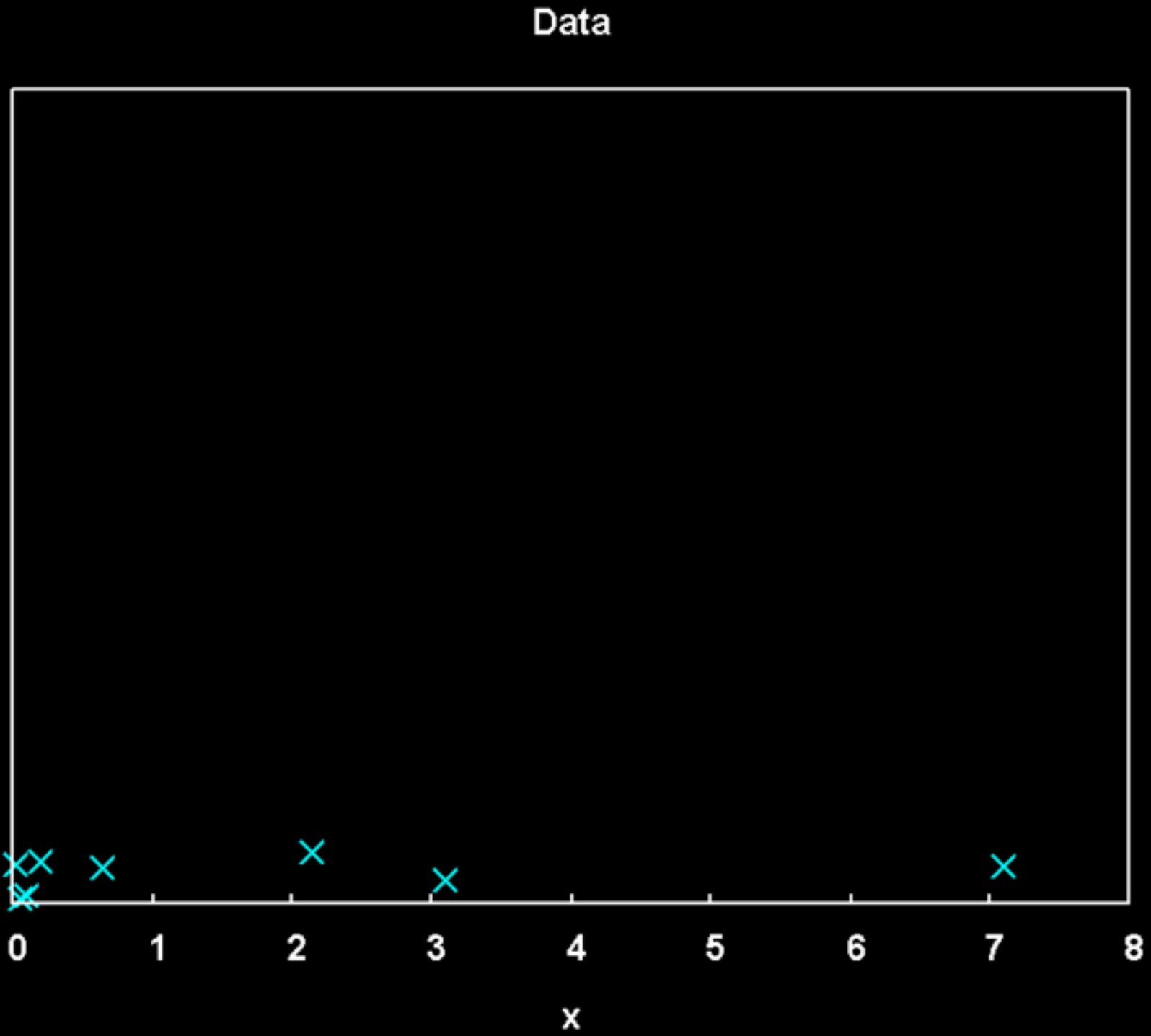

likelihood function
0.03
0.025
0.02
0.015
0.01
0.005
0
0
1
2
3
4
5
6
7
8
9
10
lambda

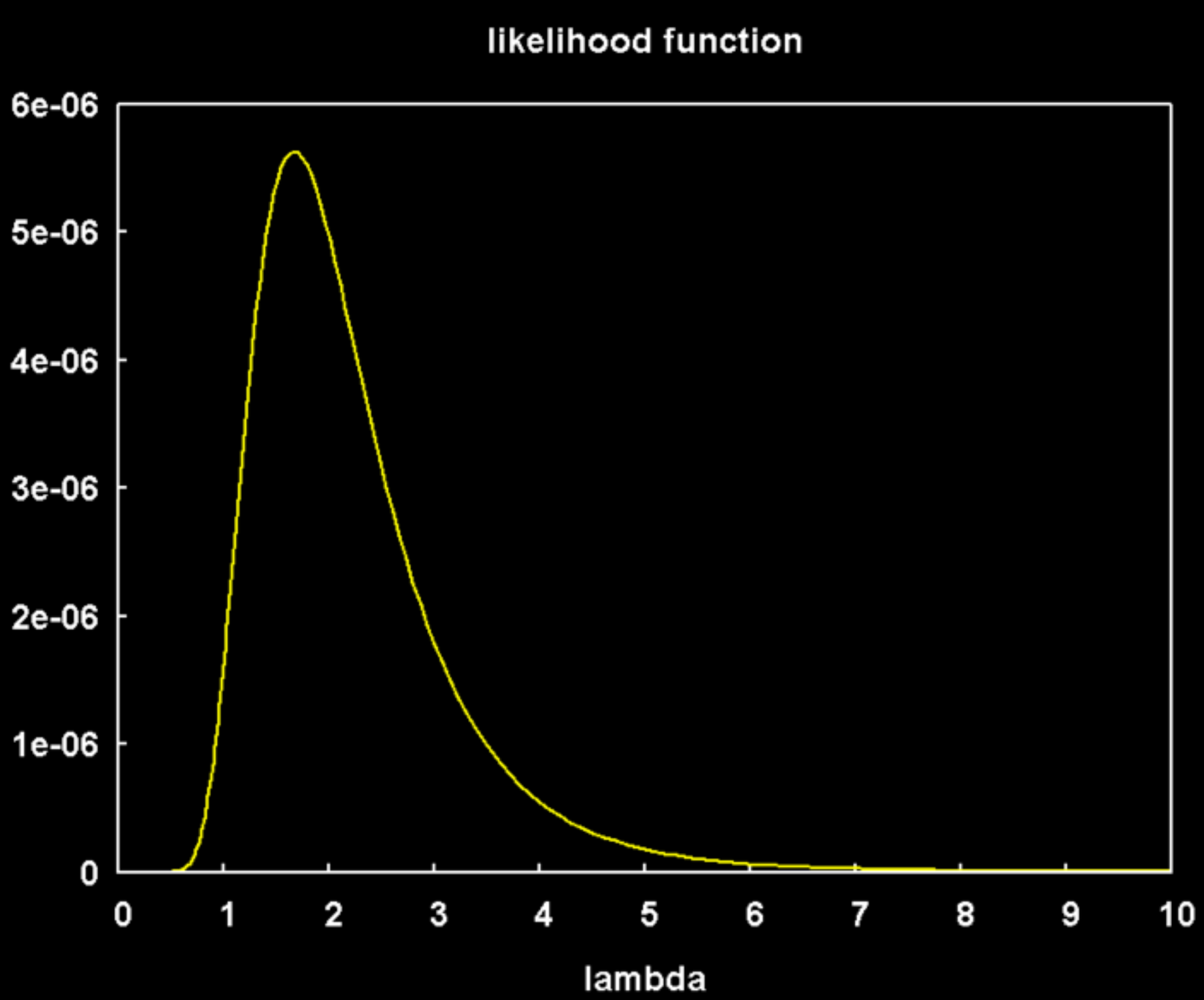

likelihood function
0.03
0.025
0.02
0.015
0.01
0.005
0
0.01
0.1
1
10
100
lambda

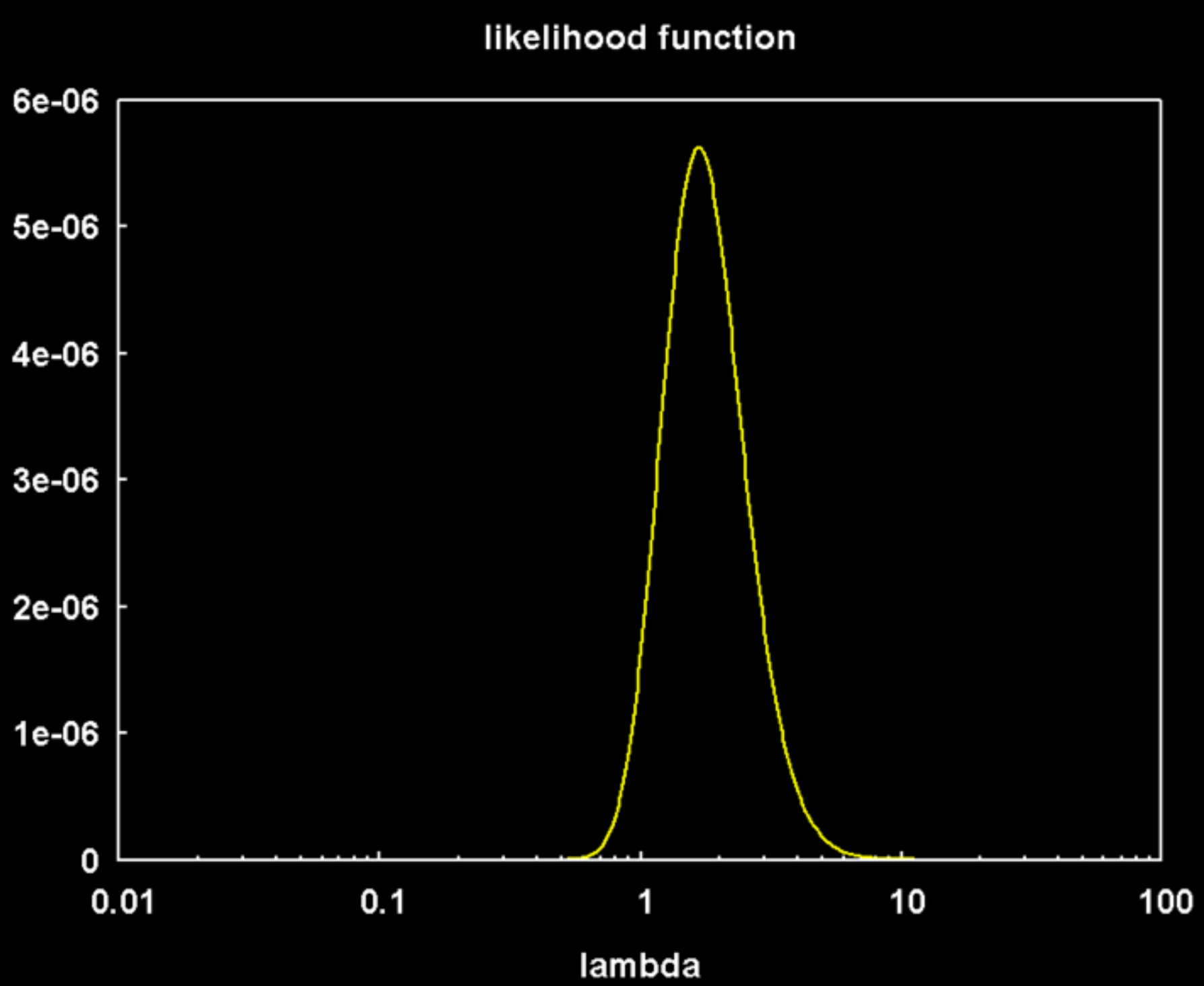

maximum likelihood distribution
1.2
1
0.8
0.6
0.4
0.2
0
0
2
4
6
8
10
12
14
16
x

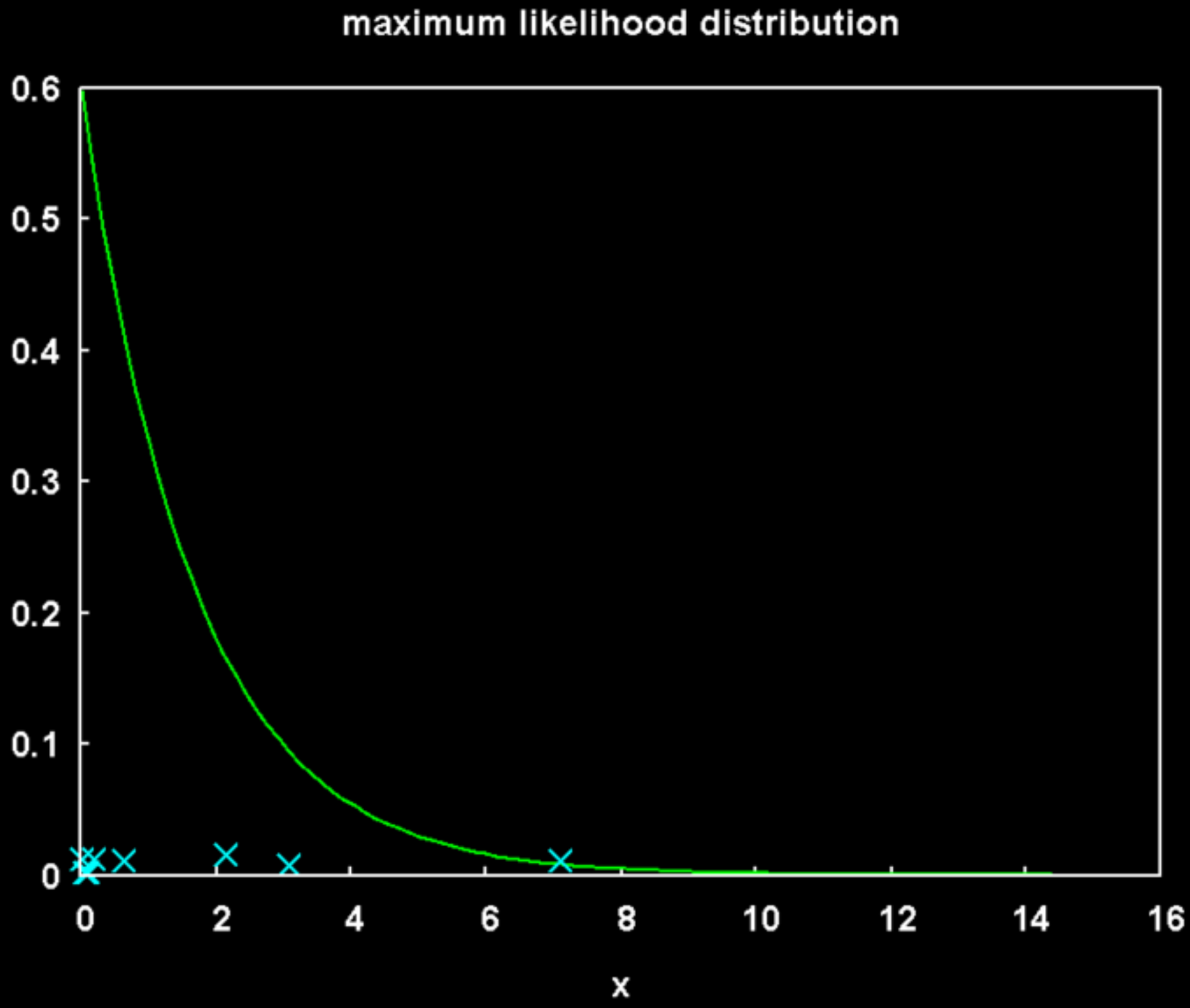

maximum likelihood distribution
1.2
1
0.8
0.6
0.4
0.2
0
0
2
4
6
8
10
12
14
16
x

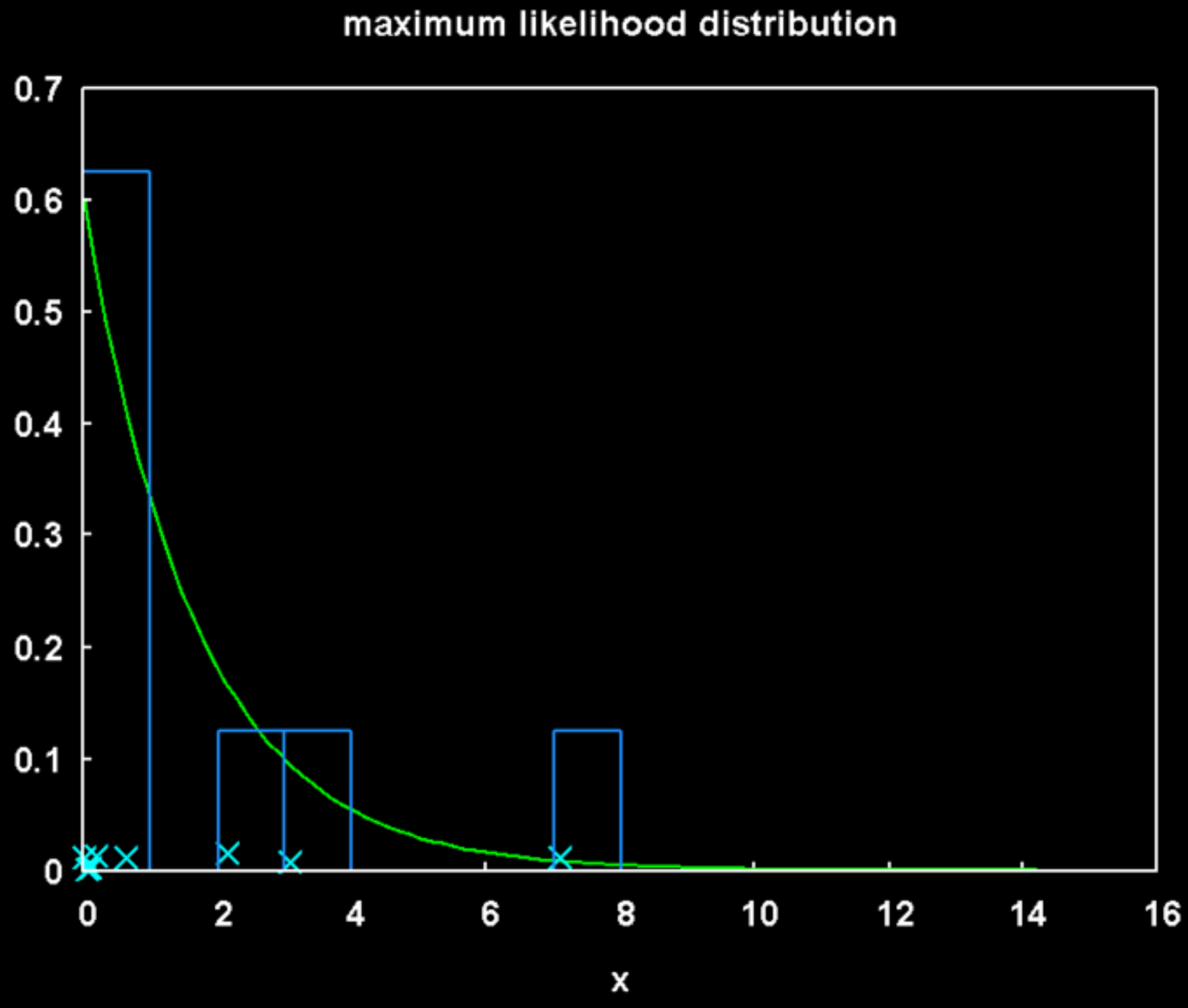

maximum likelihood distribution
0.6
0.5
0.4
0.3
0.2
0.1
0
0
2
4
6
8
10
12
14
16
x

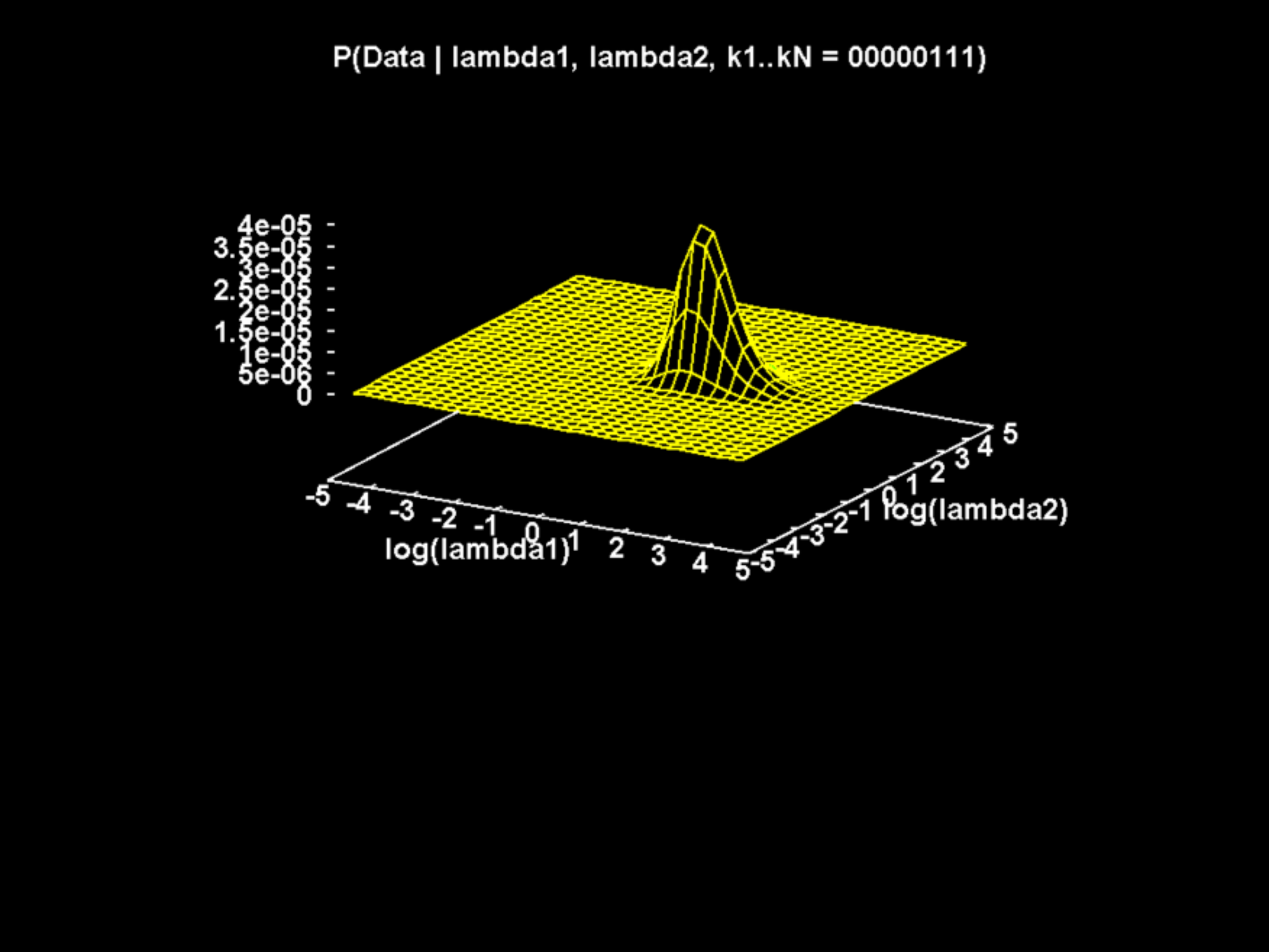

Data
0
1
2
3
4
5
6
7
8
x

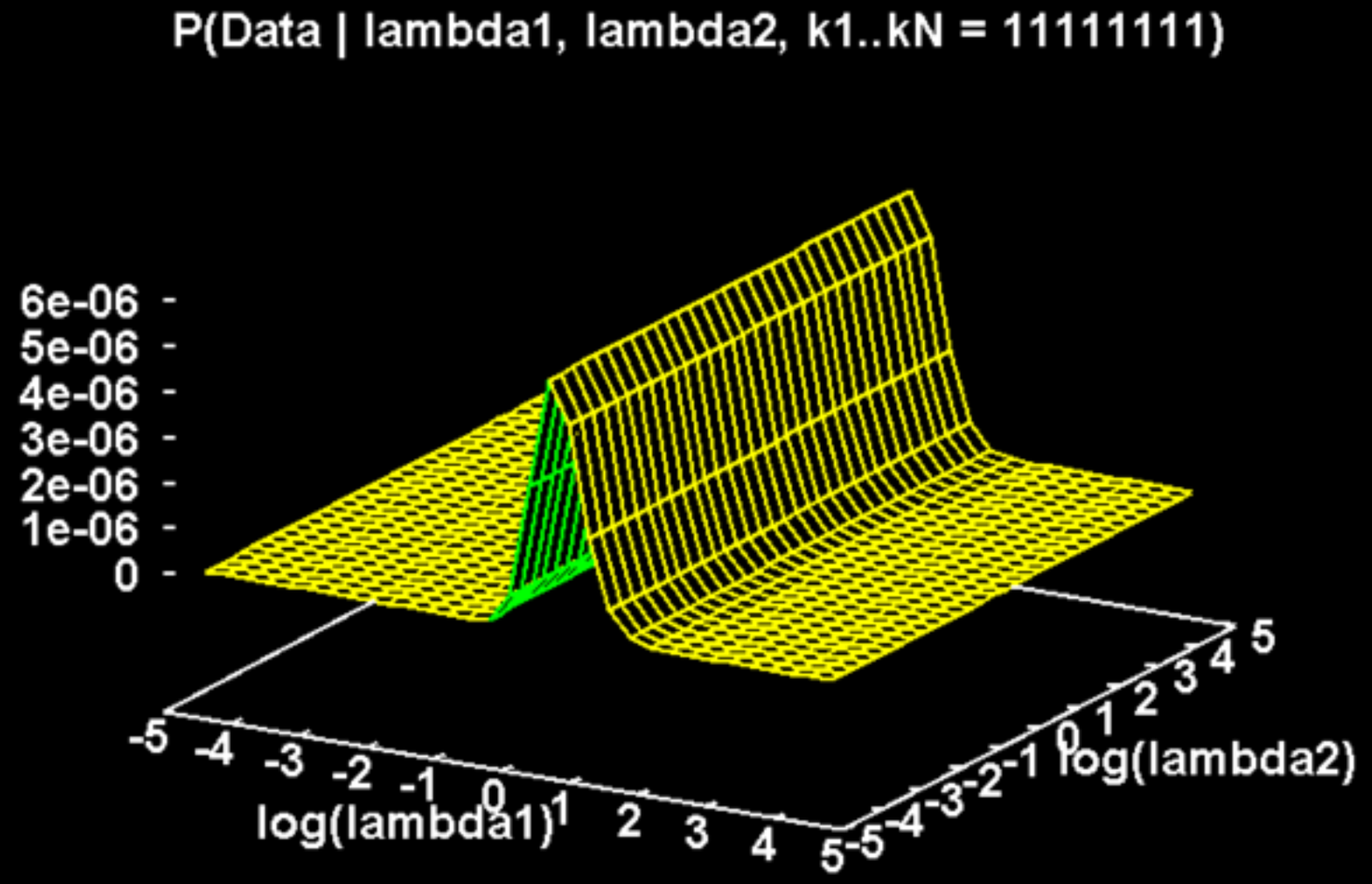

likelihood function
6e-06
5e-06
4e-06
3e-06
2e-06
1e-06
0
0
1
2
3
4
5
6
7
8
9
10
lambda

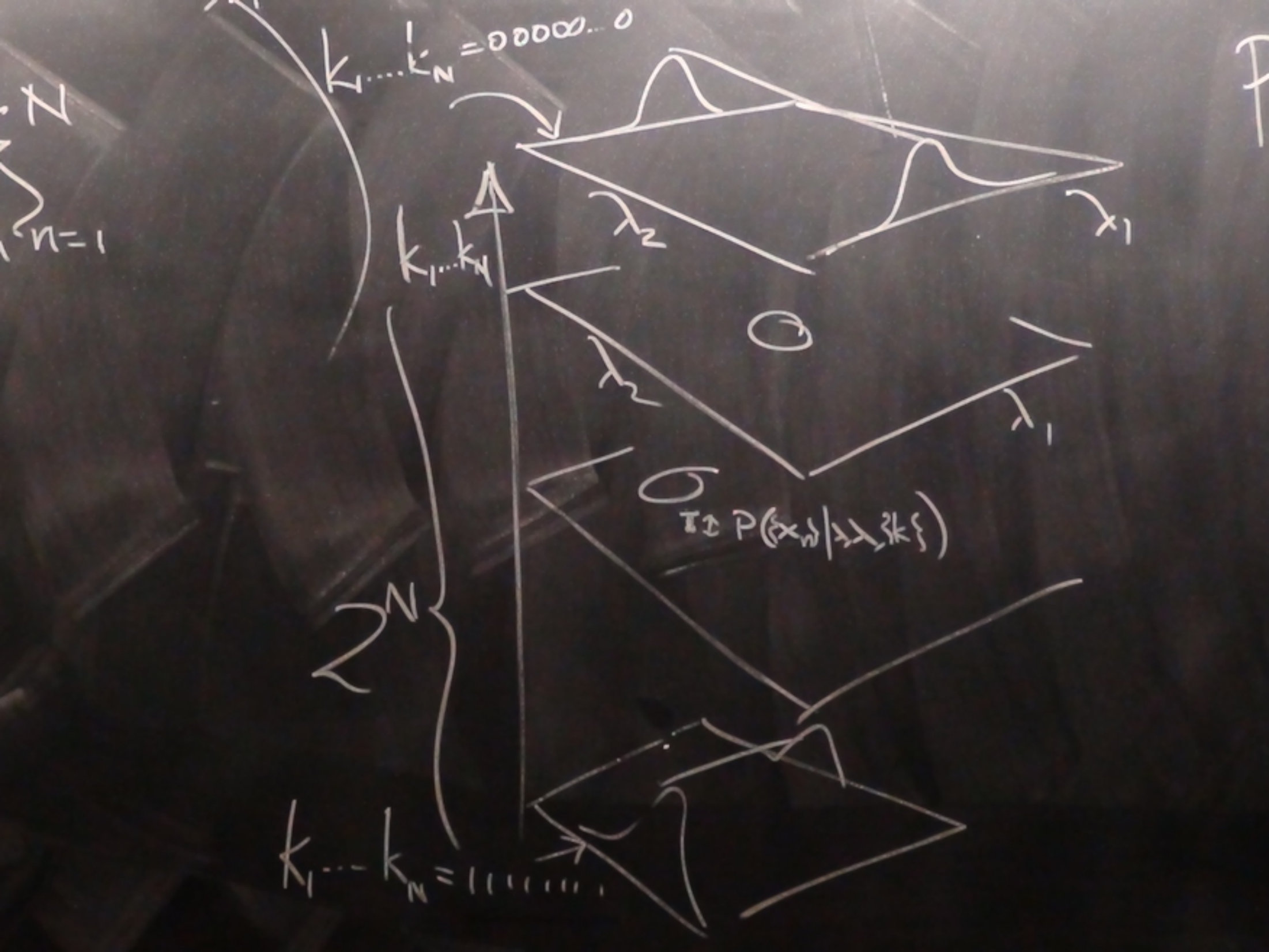

likelihood function
6e-06
5e-06
4e-06
3e-06
2e-06
1e-06
0
0.01
0.1
1
10
100
lambda

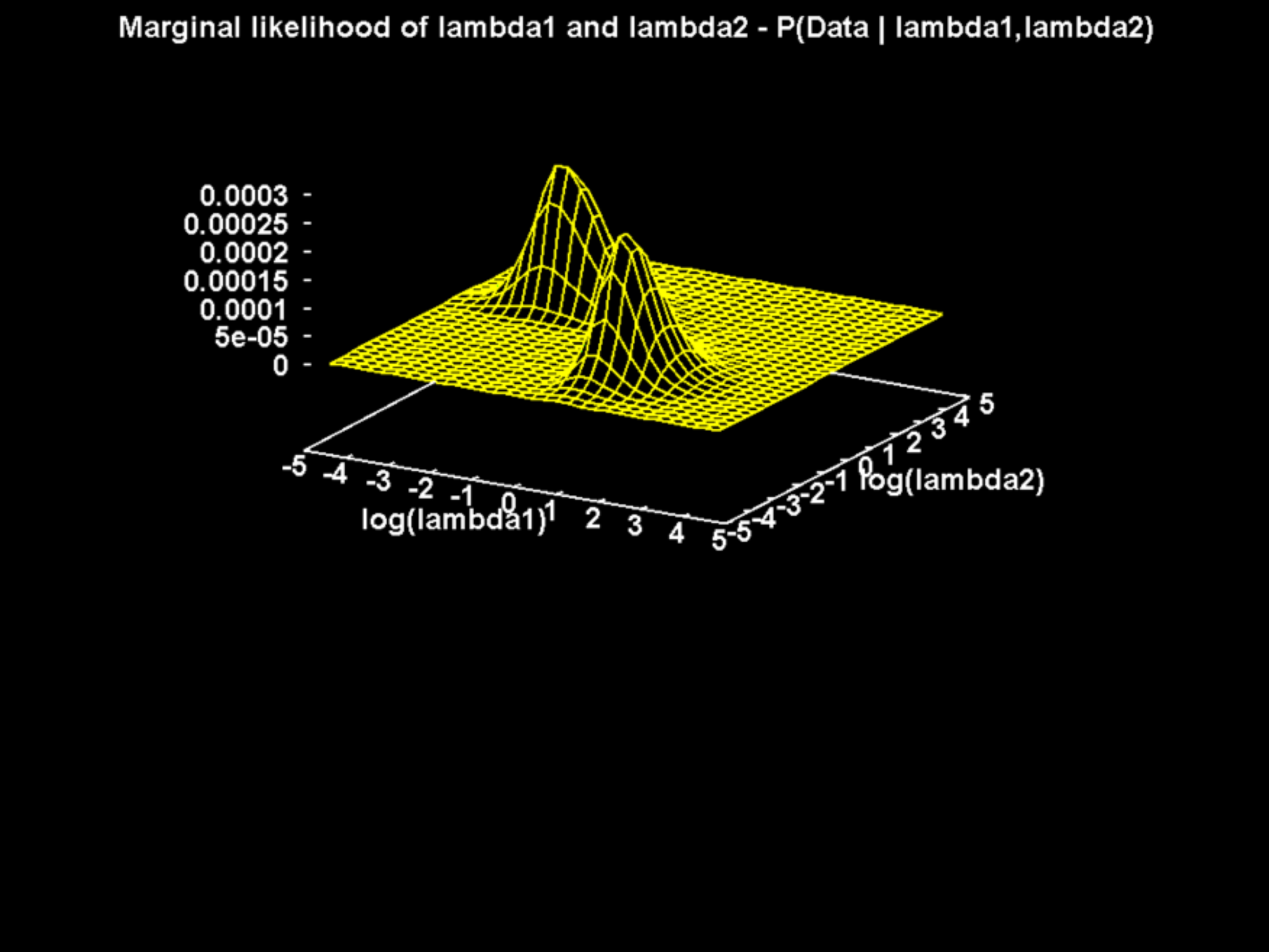

maximum likelihood distribution
0.6
0.5
0.4
0.3
0.2
0.1
0
0
2
4
6
8
10
12
14
16
x

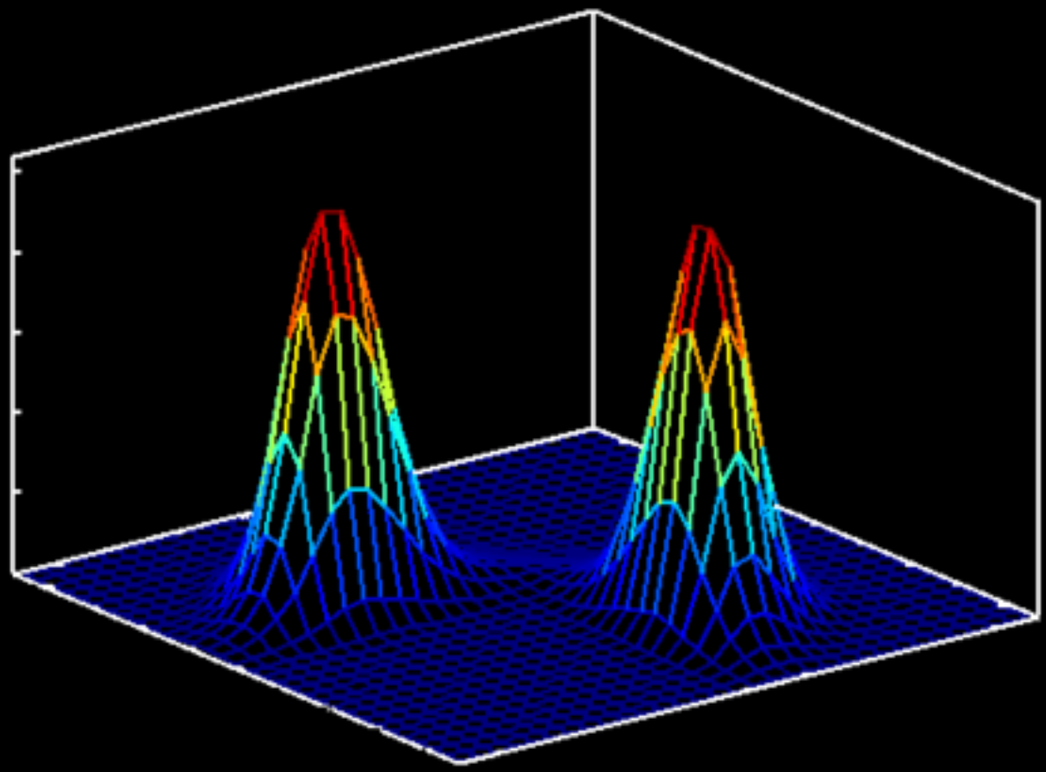

maximum likelihood distribution
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
2
4
6
8
10
12
14
16
x

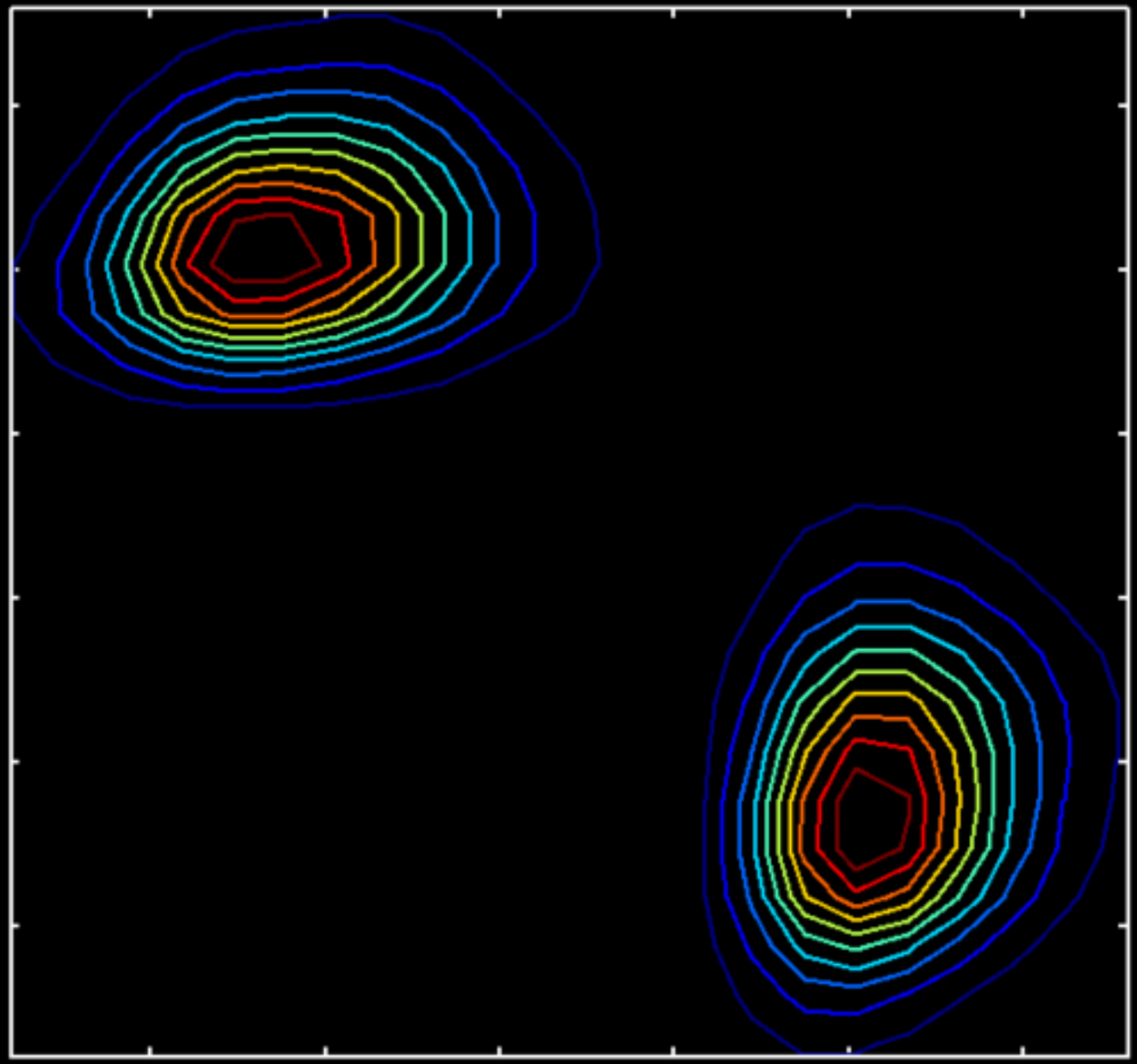

P(Data | lambda1, lambda2, k1..kN = 00000111)
4e-05
3.5e-05
3e-05
2.5e-05
2e-05
1.5e-05
1e-05
5e-06
0
-5 -4 -3 -2 -1 0 1 2 3 4 5
log(lambda1)
-5 -4 -3 -2 -1 0 1 2 3 4 5
log(lambda2)

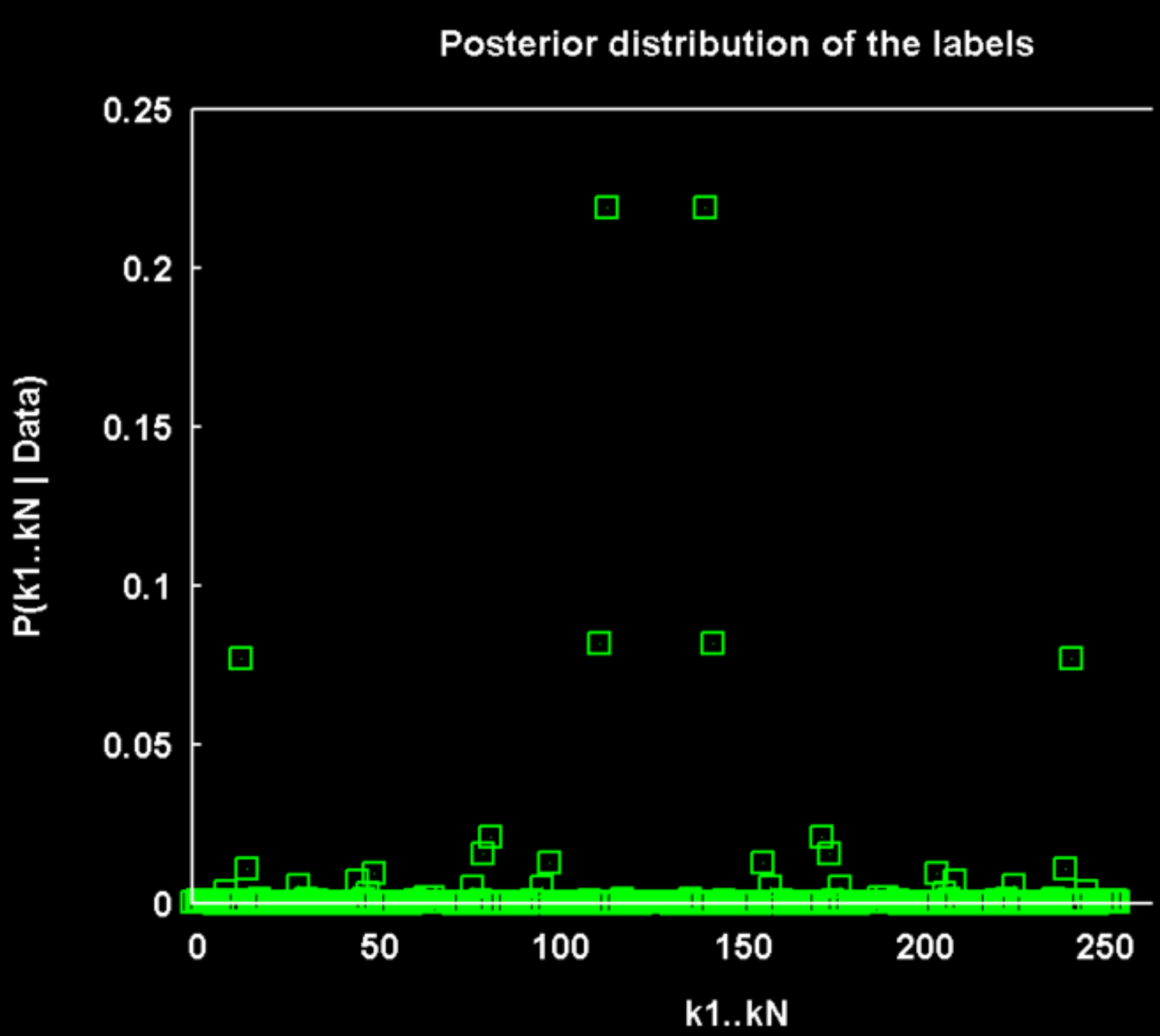

P(Data | lambda1, lambda2, k1..kN = 11111111)
6e-06
5e-06
4e-06
3e-06
2e-06
1e-06
0
-5 -4 -3 -2 -1 0 1 2 3 4 5
log(lambda1)
-5 -4 -3 -2 -1 0 1 2 3 4 5
log(lambda2)

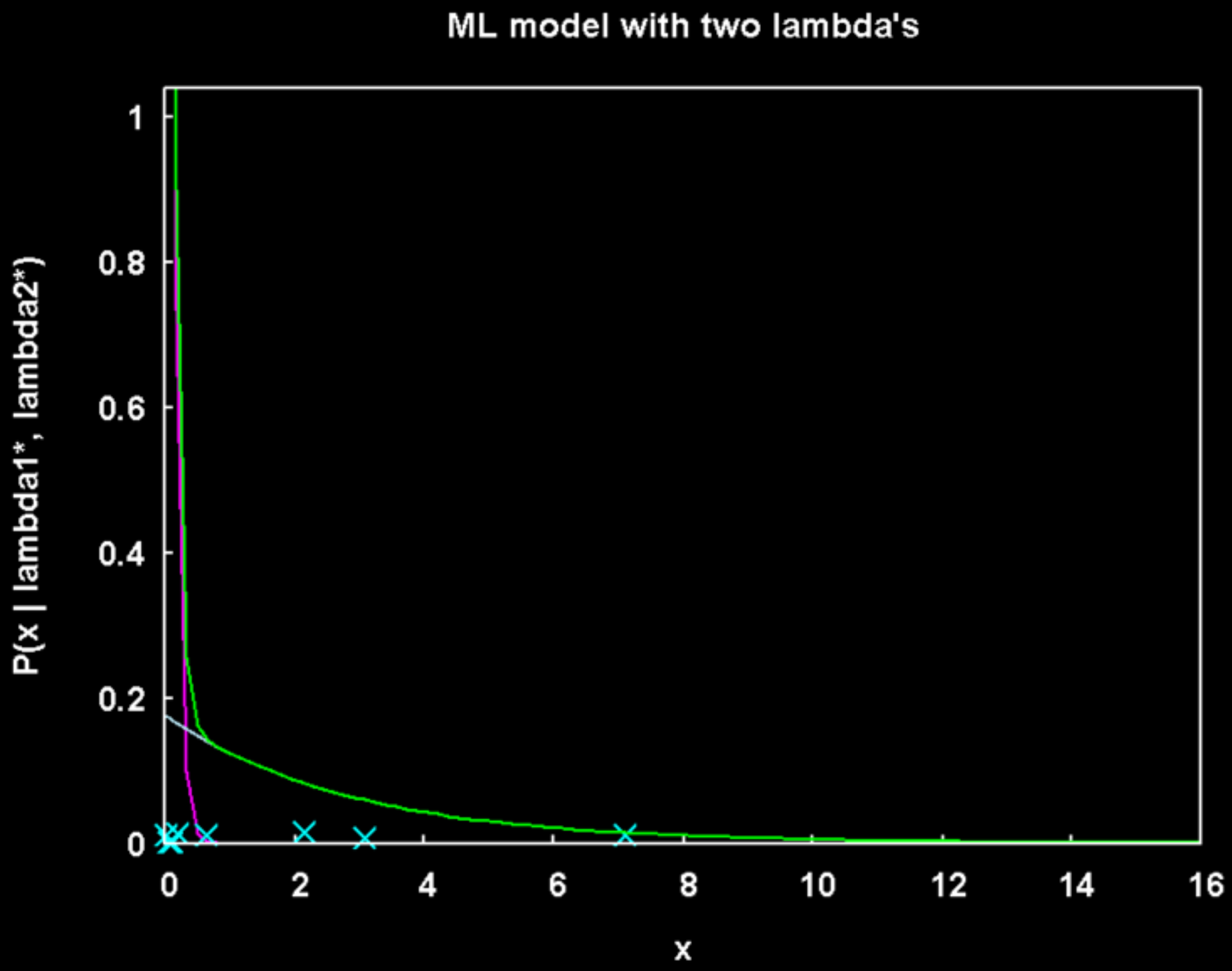

$k_1 \ldots k_N = 00000 \ldots 0$
$k_1 \ldots k_N$
λ_2
λ_1
λ_2
λ_1
2^N
$k_1 \ldots k_N = 11111 \ldots 1$


Marginal likelihood of lambda1 and lambda2 - P(Data | lambda1,lambda2)
0.0003
0.00025
0.0002
0.00015
0.0001
5e-05
0
-5 -4 -3 -2 -1 0 1 2 3 4 5
log(lambda1)
-5 -4 -3 -2 -1 0 1 2 3 4 5
log(lambda2)






Posterior distribution of the labels
P(k1..kN | Data)
0.25
0.2
0.15
0.1
0.05
0
0
50
100
150
200
250
k1..kN


ML model with two lambda's
P(x | lambda1*, lambda2*)
1
0.8
0.6
0.4
0.2
0
0
2
4
6
8
10
12
14
16
x

Data set 3 - 8 points

Data = 0.64 0.016 0.100 0.050 3.09 2.14 0.200 7.10

- Model 1

optimized lambda = 1.6637

ML_model1 = 5.6254e-06

Evidence1 = 5.4555e-07

- Model 2

Evidence2 = 1.3994e-05

ML = 2.5026e-04

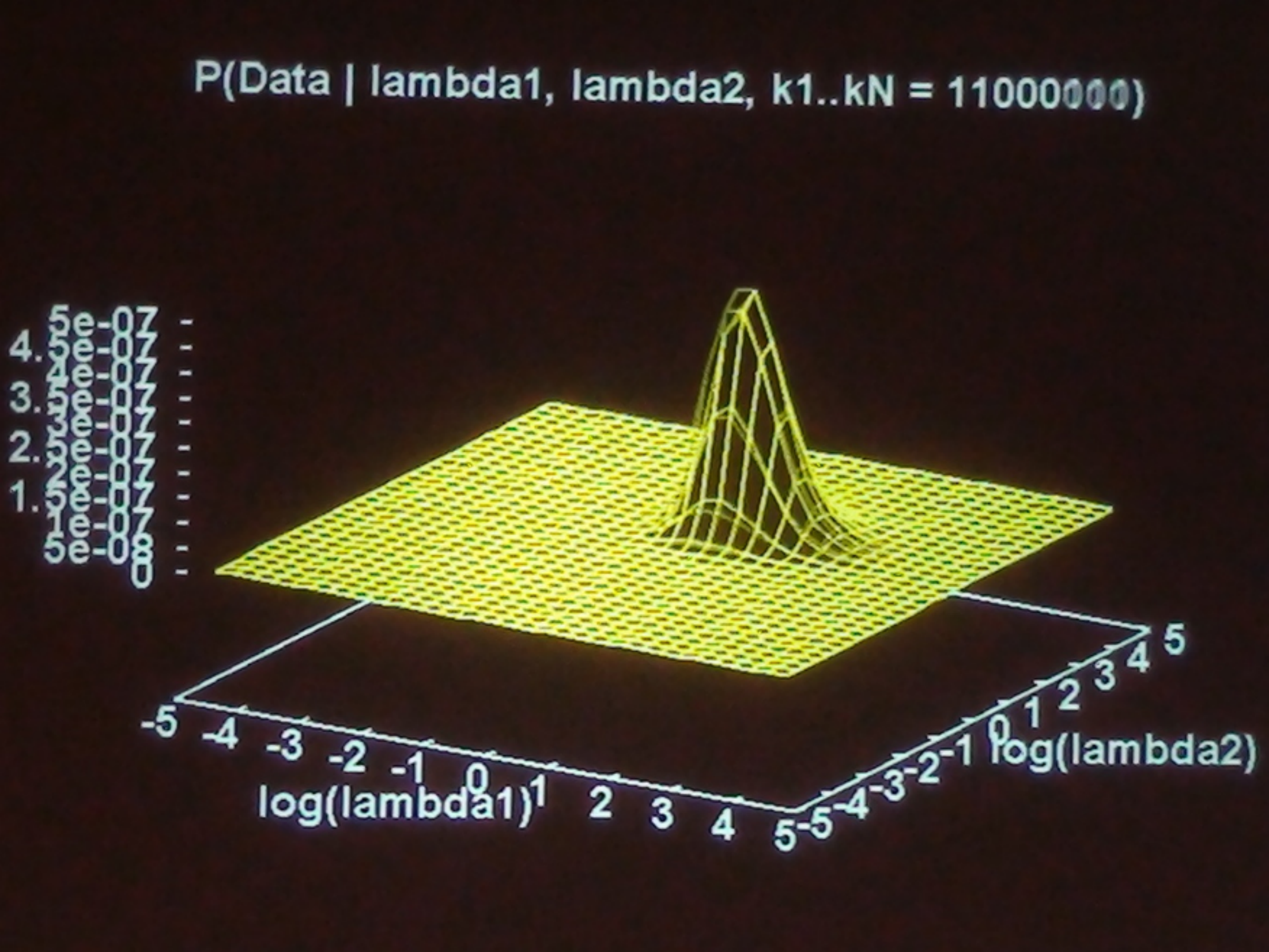
tau1ml = 0.080025 ; tau2ml = 2.8289

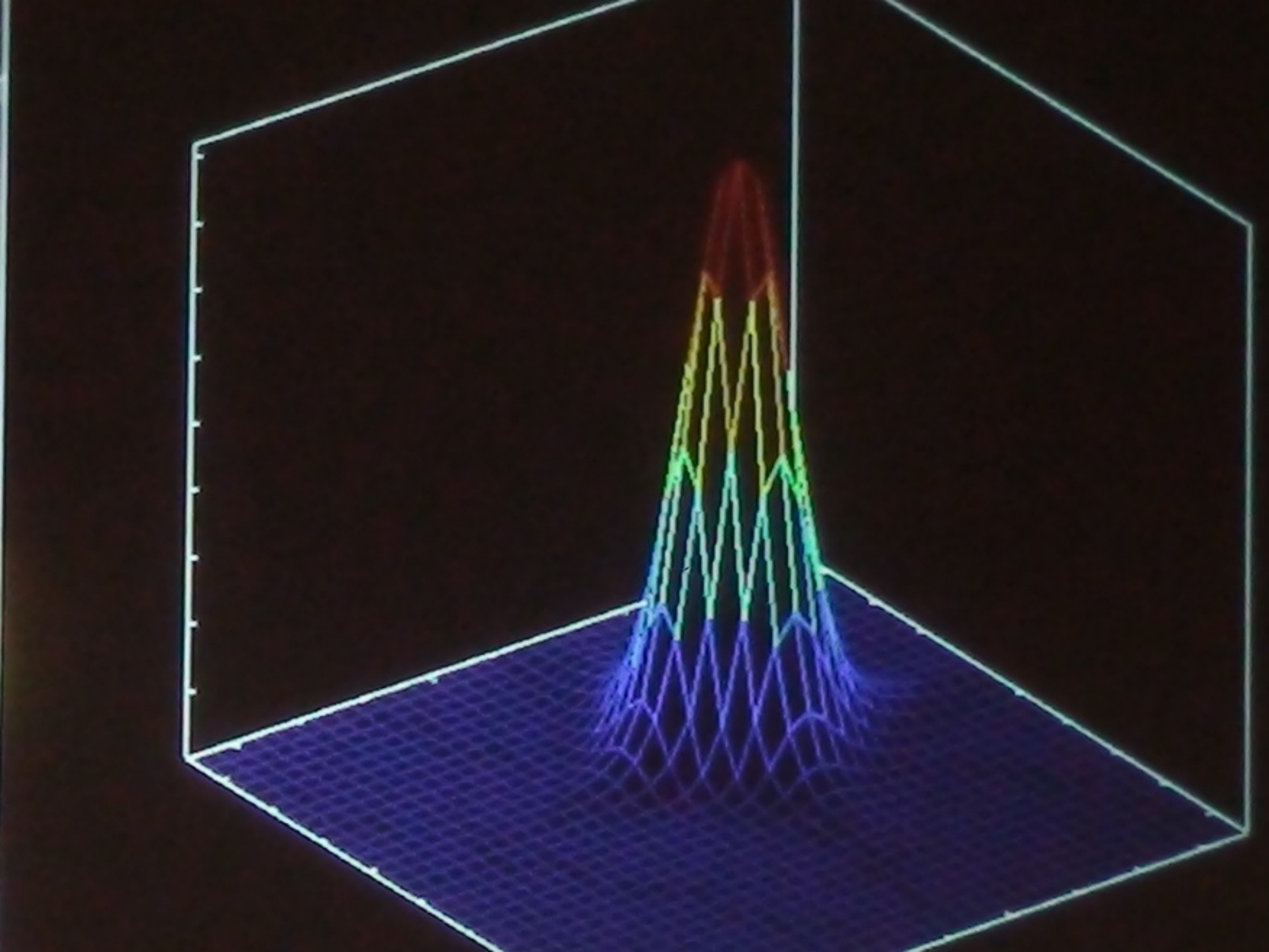
k1..kN optimized (optimal labels) = 01110010

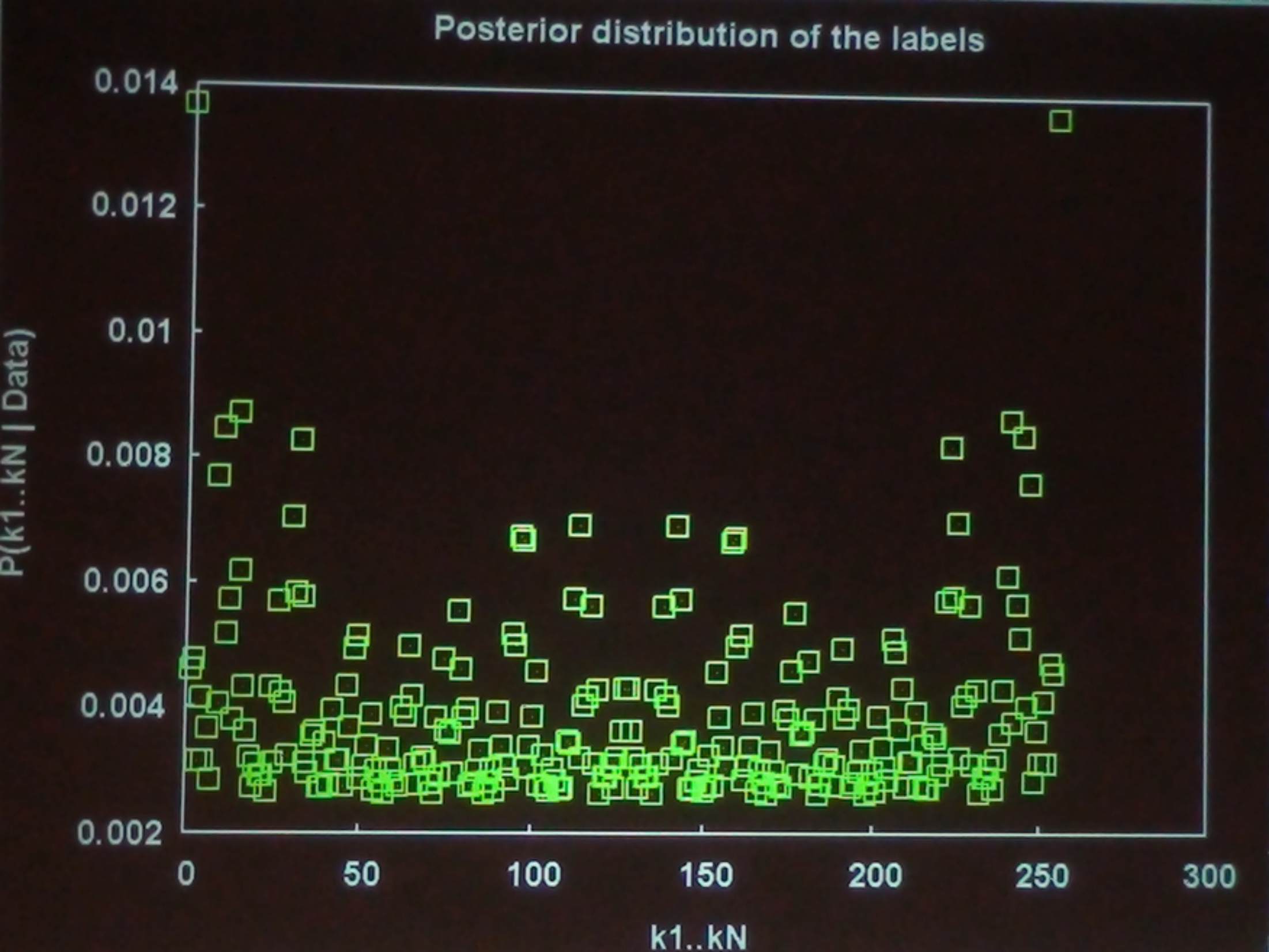
EvidenceRatio = 25.651

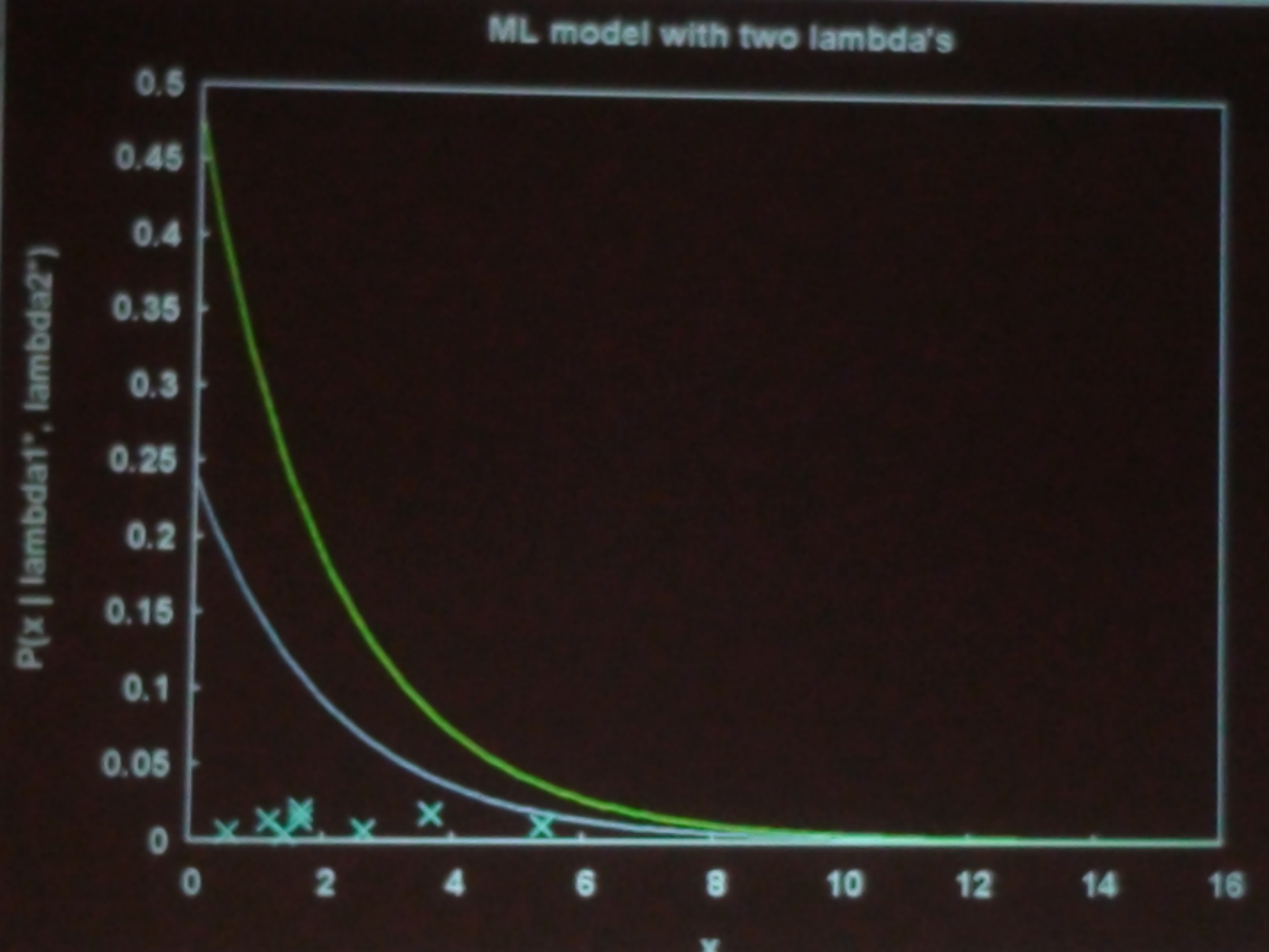
PosteriorProbOfH2 = 0.96248

PosteriorProbOfH1 = 0.037522

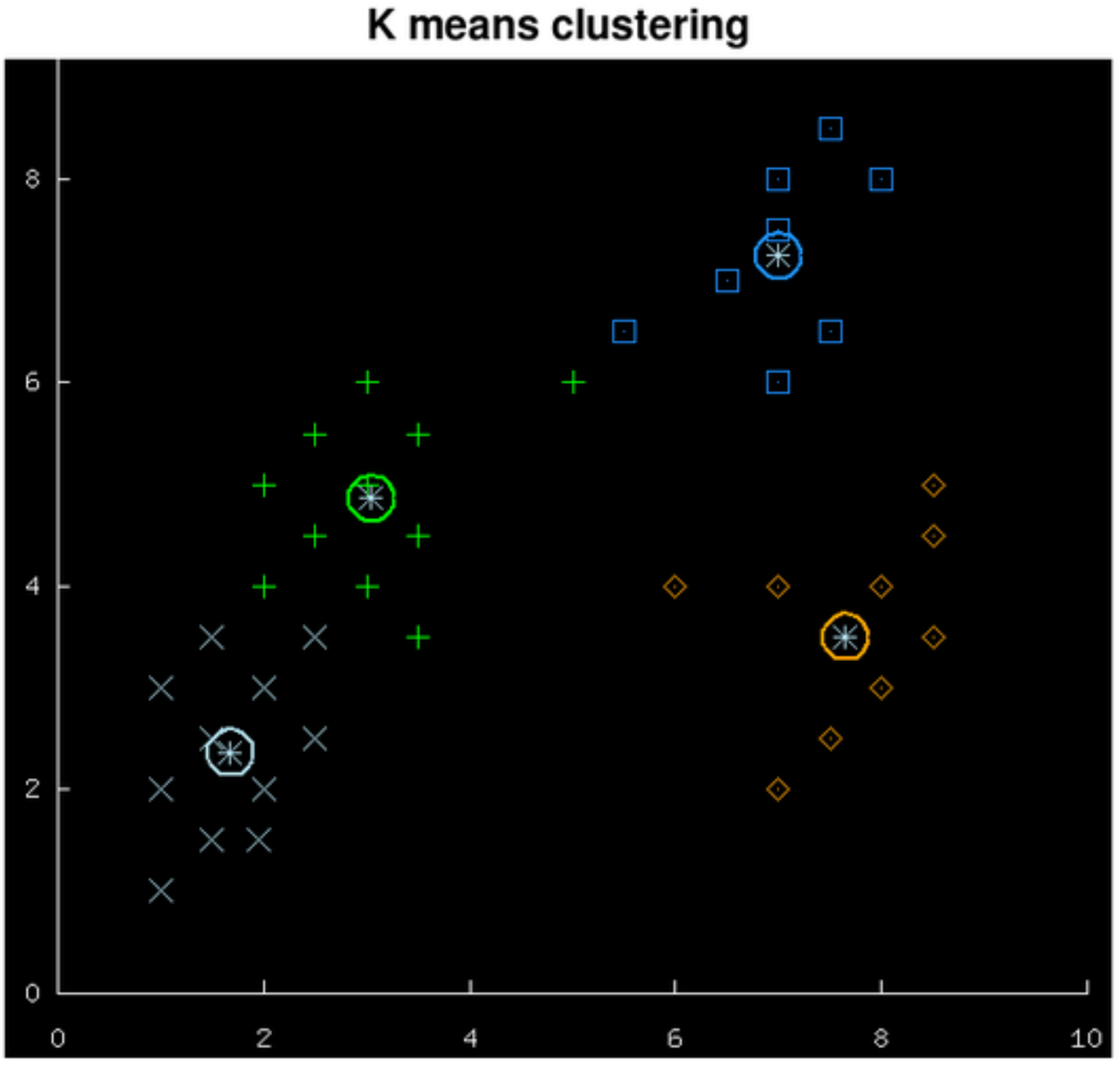

P(Data | lambda1, lambda2, k1..kN = 11000000)
5e-07
4.5e-07
4e-07
3.5e-07
3e-07
2.5e-07
2e-07
1.5e-07
1e-07
5e-08
0
-5 -4 -3 -2 -1 0 1 2 3 4 5
log(lambda1)
-5 -4 -3 -2 -1 0 1 2 3 4 5
log(lambda2)



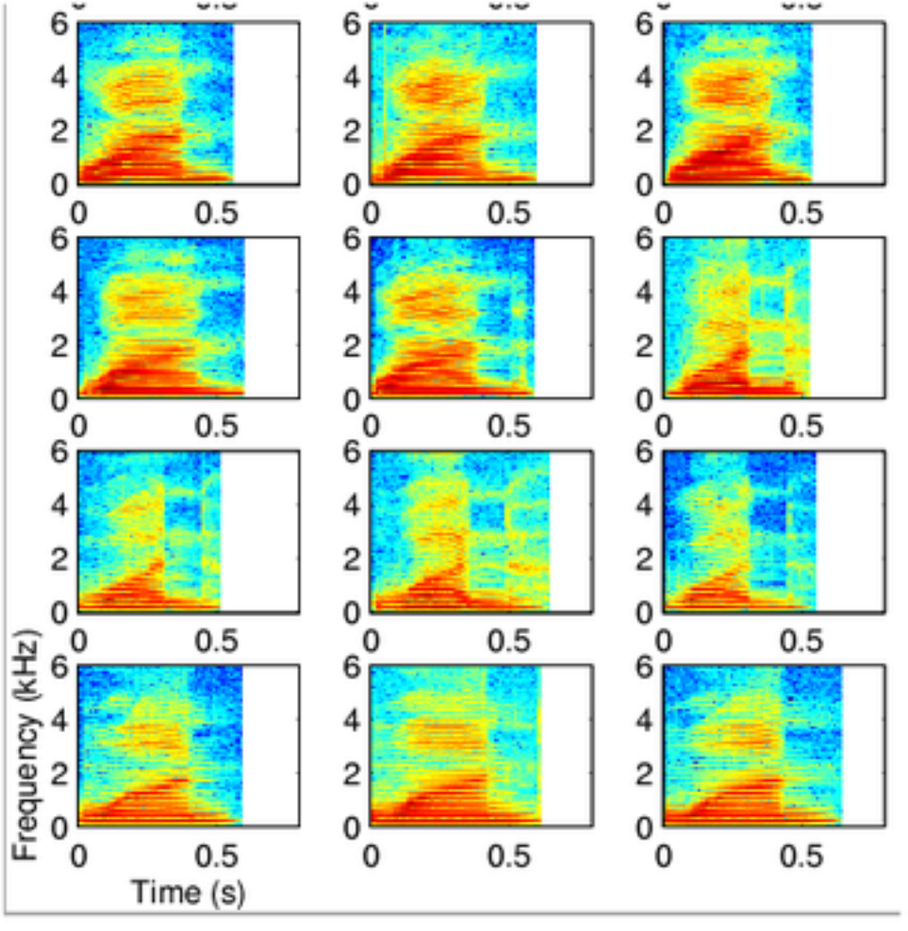

Posterior distribution of the labels
0.014
0.012
0.01
0.008
0.006
0.004
0.002
P(k1..kN | Data)
0
50
100
150
200
250
300
k1..kN

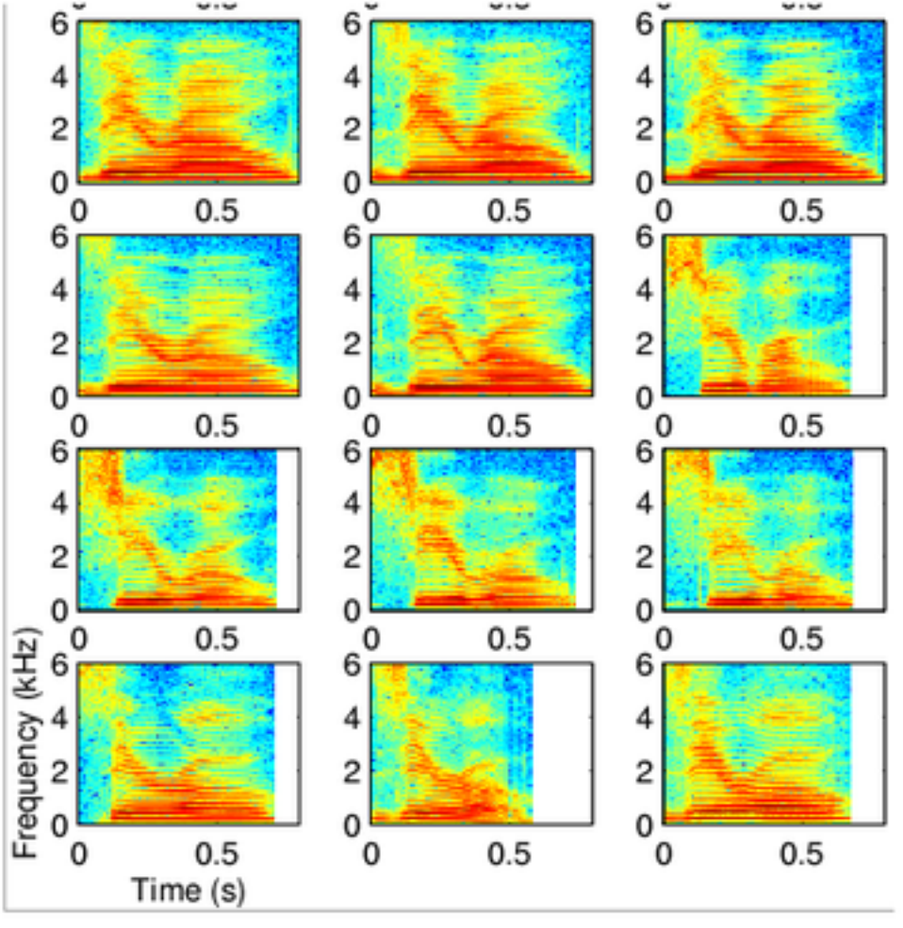

ML model with two lambda's
P(x | lambda1*, lambda2*)
0.5
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0
2
4
6
8
10
12
14
16
x

Clustering

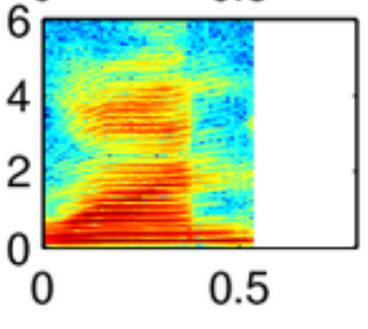


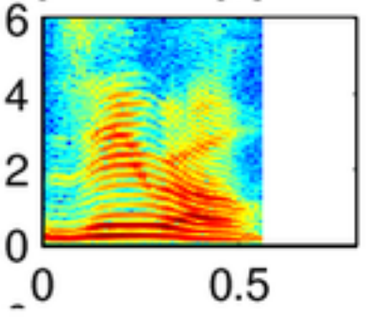
Clustering





One





Zero

Recommended homework

- Noisy channels - Chapters 8, 9, 10 (10.1-10.4 only)
  - Exercises 9.17 (p155); 10.12 (172); 15.12 (235)
  - and (if you want more practice) 15.11, 15.13, 15.15

- Invent a channel to pose to your colleagues:
  - 'what's the capacity of _this_?'

The reading associated with the current lectures is Chapters 3, 21 (especially sec 21.2), and 22 (especially sec 22.1), and 27.

Other recommended exercises are listed on handout 2.

www.inference.phy.cam.ac.uk/itprnn/